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SemaDrift: A hybrid method and visual tools to measure semantic drift in ontologies

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HIGHLIGHTS

- A novel hybrid approach to identify concepts in ontology versions across time.
- Visualizations and concept stability rankings now accessible due to the hybrid approach.
- Hybrid and existing morphing based methods integrated into end-user applications.
- Two real-world scenarios verify the novel insights otherwise hard to obtain.
- End-user evaluation with optimistic feedback.

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ABSTRACT

Semantic drift is an active field of research, aiming to identify and measure changes in ontologies across versions in time, closely related to several fields such as ontology evolution and versioning. However, practical and widely adopted methods for measuring semantic drift are mostly either not directly applied to Semantic Web formalisms or largely dependent on specific models and domains. This paper presents a novel hybrid method, which combines existing identity-based and morphing-based approaches, and, in turn, introduces further insights such as hybrid chains of concepts across ontology versions and concept stability ranking. Furthermore, it introduces the SemaDrift Application Suite, which integrates both new and existing methods, including structural and text similarity measures, provided as open-source. The applications offer a graphical user interface to calculate and visually explore semantic drift metrics, in order to support and promote their use to Semantic Web and domain experts alike. Specifically, the SemaDrift plugin for the popular Protègè platform is intended for ontology engineers, while SemaDriftFx allows a wider audience of users to visually investigate drift either as numeric output or visually through graphs. Two use case scenarios demonstrate the applicability and usefulness of the methods and tools in the domain of digital preservation and Web Services, with insights previously hard to obtain. Standard scale and concrete end-user evaluation tasks were used to acquire positive preliminary feedback on the usability of the SemaDriftFx tool.

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1. Introduction

Semantic drift, also referred to as *evolving semantics* or *semantic change*, is an active and growing area of research that observes and measures the phenomenon of change in the meaning of concepts over time [1]. With the use of Semantic Web technologies, knowledge and meaning can be encoded in ontologies. Various fields and methods study semantic drift in ontologies, such as ontology change, evolution, management and versioning [2,3], with the potential to greatly impact and benefit the multitude of domains with ontology-based applications. These mainly include the Web, e.g. the Linked Data best practices of interconnecting published data [4], and various others, such as cultural heritage, digital preservation and bioinformatics.

Several ambiguous terms are entailed in the field, bearing slightly different meanings. Semantic drift usually refers to ontologies or other formalisms relying on Semantic Web technologies. Concept drift may refer either to the drift of concepts within ontologies but also to concepts or else target labels in machine learning, referring to an entirely different field. Other such ambiguous terms include semantic shift and semantic decay [5–7].

Similarly to semantic drift, ontology evolution is the research field which studies the process of an ontology changing in terms of size, content and management in order to accommodate dynamic

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changes and knowledge interchange in industrial and academic applications [6]. This phenomenon mandates the need for an efficient monitoring and management process. However, generally applicable and widely adopted methods and tools to measure semantic drift are mostly either not directly applicable to content encoded with Semantic Web formalisms (RDF, RDFS, OWL) or largely dependent on specific models and domains.

Addressing this shortcoming, the paper introduces new methods and tools to measure semantic drift in ontologies. Initially, it presents a background study disambiguating terms and clarifying existing approaches regarding change, drift and shift measurement. Then, a framework for measuring semantic drift across multiple versions of ontologies is presented. We build upon existing work in the field of concept drift [7], which established the terms and metrics of the two main methods: the morphing-based and the identity-based approaches. In the former, the identity, i.e. concept correspondence across ontology versions, is known, while in the latter it remains unknown. This paper extends those methods, implementing them within an open and reusable software solution. In addition, an alternative, hybrid approach to measure drift is introduced, exploring the ability to infer the correspondence of concepts, rather than have it explicitly entered (identity) or left unknown (morphing-based). As a result, the framework provides a method to monitor evolving semantics as a vehicle to measure and manage ontology change. Furthermore, two realistic use case scenarios validate the usefulness of the tools and methods with previously hard to obtain insights that enable scholars to detect interesting areas of drift. The two scenarios involve existing ontologies, one in Digital Preservation, where identifying long-term changes is crucial, and the other involving a standardized Web Service ontology. User evaluation also shows positive feedback.

With regards to previous research in the domain of measuring semantic drift in ontologies, the paper extends established methods. The implementation of the morphing-based method was initially introduced in [8]. The first software tool with a Graphical User Interface (GUI), was introduced in [1], presenting a plugin for the Protégé platform with the basic functionality, to compare two ontology versions and display the numeric outcomes of the metrics. The work in [9], further validated the usefulness and generality of the methods across domains, by presenting two realistic scenarios with detailed comparisons between the morphing-based and the novel hybrid approach, both via the proposed tool.

On top of existing work, this paper introduces a hybrid approach to measure drift, which allows not only to detect concept identity, but also to produce previously inaccessible graphs and stability rankings. Tools that integrate the new methods with a GUI are also introduced in this work, intended both for experts and non-experts alike. The tools offer novel functionality, such as building a series of ontologies (ontology versions, usually in chronological order) and evaluating the metrics, both pairwise and over the entire series. Most importantly, the new visualizations now aid the users to further assess and investigate drift. Specifically, both hybrid chains via the hybrid approach and morphing chains via the morphing-based approach, can be built and optimized according to preferences. SemaDriftFx, the most advanced of the proposed tools, is validated through two realistic use cases, showing how valuable insights are obtained without much knowledge and manual work. In one case, the tools are used to investigate drift in a real taxonomy of digital preservation of media art, with instances transcribed from actual notes by domain experts, provided in the context of the PERICLES FP7 project.[1]

The paper is structured as follows: a background study on semantic drift is presented in the next section, so as to disambiguate relevant terms. Section 3 presents related work, of studies that aim to provide means and tools for measuring drift. Section 4 presents the proposed SemaDrift metrics and methods, followed by Section 5 describing the proposed application suite that incorporates them into tools for developers, experts and non-experts alike. Section 6 presents diverse use cases that validate the usefulness and the validity of the tools, while Section 7 presents a preliminary end-user usability evaluation. The paper concludes with Section 8, which discusses lessons learned and Section 9 with conclusions and directions for future work.

2. Background

This section presents an overview of the scientific background in the field of studying semantic drift. Moreover, it is focused on disambiguating the various terms that are used to describe semantic drift, as previous works use them. The terms are often used interchangeably, as synonyms, but in some cases they might imply something different in meaning. In an effort to disambiguate the terms, the following subsections present the most popular definitions found in literature, while the final subsection tries to draw relationships among them.

2.1. Semantic drift

Semantic drift refers to how the features of ontology concepts gradually change, as the underlying knowledge domain evolves. Alternatively, semantic drift refers to the ability of concepts to be re-interpreted by different user communities or in a different context, introducing a risk for them to lose their rhetorical, descriptive and applicative power [10]. According to [10], it can also be defined as the gradual change of a concept's semantic value as it is perceived by a community. It can be characterized as intrinsic or extrinsic, depending on whether a concept's semantic value is changed with respect to other concepts in the ontology or to the phenomena it describes in the real world.

Drift can also be classified as either non-collective, inconsistent or consistent collective [11]. If a concept is exposed to extrinsic, but not intrinsic drift, it means that the whole ontology is undergoing a consistent, collective drift that may not necessitate any changes to it. On the other hand, no extrinsic drift together with substantial intrinsic drift means that relationships between concepts may no longer be valid, even though the meanings of the concepts themselves have not changed. In cases of both extrinsic and intrinsic drift we are dealing with inconsistent collective drift rendering the ontology no longer valid. Overall, semantic drift is the general case of gradual transformation of a concept in semantic constructs. This work aims to investigate exactly those transformations at any level and, therefore, semantic drift is the dominant term adopted.

2.2. Semantic change

Semantic change refers to the extensive revisions of an ontology, and can, therefore, be associated with versioning. Semantic change occurs when the internal structure of a concept in two ontologies is different [12]. An isomorphic change refers to the structure being unchanged while the names might have been altered. Other researchers consider semantic change to occur when an ontology revision presents so many alterations that it can be considered as an entirely new one [2]. In all cases, the ontology authors are responsible to decide whether this will occur or the ontology will continue to represent the same conceptualization. All in all, semantic change can be considered a specific case of drift, which is so large that the meaning of the ontology has actually been altered.

2.3. Concept drift

Concept drift is defined as a change in the meaning of a concept over time, possibly also across locations or cultures. It often refers

[1] http://pericles-project.eu/.

to a problem in the field of data mining and machine learning, when learned models lose their predictive power over time [13]. In this case, semantic and concept drift might be referring to two entirely different fields. To be exact, concept drift in data mining means that the underlying distribution of data is changing in unforeseen ways, making target variables hard to predict, while semantic drift is the language-related version of the same phenomenon [10]. However, previous works have begun bridging the gap between the two fields, by applying concept drift notions from data mining (such as label, intension and extension) in semantic drift as means for measuring it. This consideration of three types of drift: concept label drift, intensional drift and extensional drift has been adopted also in [14]. In the context of this work, the meaning of the term in the context of data mining is excluded. The term from now on refers to the transformation of meaning of an ontology's underlying concepts between versions.

2.4. Concept shift

Concept shift refers to the subtle changes in the meaning of related concepts over time. It can be studied by using chains of extensional, i.e. instance-centric, mappings that represent those subtle changes [7]. Concept shift often occurs in the course of evolution, so that the actual meaning of concepts better represent the structure of the real world. While some shifts of concept meaning are performed explicitly, they can also be implicit, through changes in other parts of the ontology, e.g. in properties [6]. The term "topic shift" can also describe the same phenomenon [13]. Overall, concept shift can be considered as a more specific case of drift, where meaning has migrated from one concept to another.

2.5. Concept change

Concept change refers to the broad variety of adaptations and alterations that can occur for a concept in an ontology. Such changes can be either conceptual (e.g. changing concept relations), specification or representation [3]. Concept change can be tracked by investigating obsolete concepts that have changed their name but maintained their identifiers and a history of changes that can later be examined [15]. In parallel to semantic change, concept change is a specific case of concept shift, so large that the meaning of the concept has actually altered.

2.6. Semantic decay

Semantic decay refers to the reduction of semantic richness of concepts. The volume of facts that can be inferred from a concept within the context of Linked Data and a particular dataset has been proposed as a measure of this richness. By using this metric, it has also been proved that the more a concept is reused, the less semantically rich it becomes [5]. In perspective to this study, semantic decay is a specific case of semantic drift where semantic richness of concepts has been decreased.

2.7. Ontology versioning

Concept or ontology versioning refers to building, managing and providing access to different versions of an ontology [3]. Another definition is that versioning methodology provides users of the ontology variants with a mechanism to disambiguate the interpretation of concepts [2]. It is also linked (but not identical) to ontology evolution by the ontology and database engineering communities, as both research fields aim to represent change and handle different variants of ontologies [3]. However, one of the differences between them is that ontology evolution concerns changing an existing ontology while maintaining consistency, whereas ontology versioning follows a "copy-first" strategy where changes are effected in a new, duplicate version of an ontology [16]. As a result ontology versioning is a more inclusive research area than semantic drift, concerned not only with identifying the changes within, but also with the management and access of the versions.

2.8. Overview of terms

The above term definitions according to literature help disambiguate the various terms in the topic. The previous subsections also placed them in relation to the most general term adopted in this study, semantic drift. In this subsection, the review considers works in the fields of Semantic Web and Linked Data. Relationships and links among them are materialized through their actual shared usage in literature. To draw these relationships we treat each term as a different research topic. For each topic we list the number of works that focus on it, either directly or indirectly, along with the meaning they attach to it. Furthermore, we consider that an existing study directly targets a topic when it provides definitions and methods for measuring it explicitly, and indirectly when it mentions it as a secondary target or simply as a synonymous term for the main, direct target. The outcome is depicted in Table 1, where checkmarks indicate a direct focus to a topic and checkmarks in parentheses an indirect or secondary focus of a work to a topic.

While on the table the works and their respective focus are clearly apparent, their relationships are not so evident. Fig. 1 displays a graph generated from that table, making the relationships directly visible. In the graph, each term or field relevant to semantic change is displayed as a large node. Each term is linked with smaller nodes, depicting the studies that refer to it either directly (in color) or indirectly (in gray). While this state-of-the-art study is not complete, emerging relationships at least in those popular works of research are emerging. As illustrated, concept drift, concept change and concept shift are closely related, but still connected to semantic drift, while semantic decay is a term less connected to the rest. Additionally to the relationships between fields, their penetration and adoption in research is also apparent. For instance, the most popular terms are concept change, concept drift and concept shift, with the least popular being semantic decay, topic shift and topic drift.

According to the definitions and relationships of the fields discovered so far, the rest of the paper focuses on semantic drift as a primary target, namely to describe changes in the concepts of ontologies across versions in time. While concept shift is quite close, it usually refers to concepts as constructs outside an ontology, and established methods exist to measure it, such as those in [13]. In fact, we aim to extend methods the methods in [13] and provide the means to apply them in concepts in ontologies, bridging the gap between the theory and practice of semantic and concept drift.

3. Related work

The problem we address in this paper is to provide methods and tools that are directly applicable (i.e. with minimum data transformations and manual intervention) to ontologies in order to measure and visualize semantic drift in them. This section examines existing work that aims to provide means to address this problem.

To begin with, the change of word meaning, i.e. semantic change, has been thoroughly studied in the field of linguistics. [18] have discovered statistical laws of semantic change by extracting features of words and mapping them to vectors, (word embeddings). [19] have developed a dynamic Bayesian model of meaning change across time, which infers temporal word representations as a set of senses and their prevalence. Unfortunately, the works in linguistics cannot apply directly for measuring semantic drift

Table 1
Works related to Semantic Web or Linked Data and their direct or indirect (in parentheses) focus to a topic using a certain term.

Paper/Terms	Semantic change	Semantic drift	Concept drift	Concept shift	Concept change	Concept versioning	Semantic decay	Topic drift	Topic shift
[12]	✓								
[13]		✓	✓	(✓)					(✓)
[17]			✓		(✓)				
[15]					✓				
[14]			✓		(✓)				
[7]			(✓)	✓	(✓)			✓	
[10]		✓	✓		(✓)				
[3]					(✓)	(✓)			
[2]	✓					(✓)			
[11]	(✓)	✓							
[6]				✓					
[16]						✓			
[5]							✓		

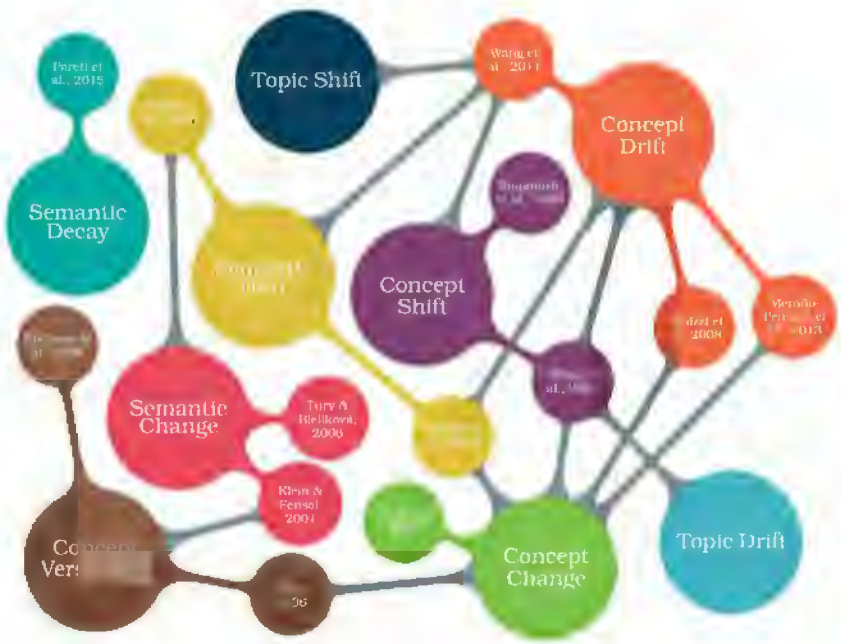


Fig. 1. Notions and terms around semantic drift, linked through literature.

in ontologies. While the former approach extracts the reasons of change, the latter models changes of word meaning across time. Like most linguistic approaches, they focus on text and not on Semantic Web formalisms. This means that, in order to be transferred over to this domain, the approaches would require semantic representations of words, their properties and relationships.

Measures of semantic richness of Linked Data concepts have been investigated in [5], proving that increasing the reuse of concepts decreases semantic richness and, thus, causes semantic decay. Other studies have examined change detection between two ontologies at a structural or content level [12]. Concept drift has been measured either by clustering while populating ontologies [17] or by applying linguistic techniques on textual concept descriptions [11]. A vector space model by random indexing has been utilized to track changes of an evolving text collection [10]. An effort not to detect but drive the process of change is presented in [6]. However, we deem some of these techniques not directly applicable to ontologies as they regard other types of data, such as text. Other techniques that involve content are also not guaranteed to be applicable either, as some ontologies do not contain enough content (instances) or even not at all.

One of the solutions that we deem most appealing to our problem transfers notions applied in machine learning concept drift to semantic drift [13]. The notions of label, extension and intension are not only introduced, but are also further defined in ontology terms, which renders them directly applicable in the Semantic Web context. In detail, label drift refers to class descriptions in text, intension refers to properties and extension to class instances. Building upon previous work, [7,13] introduce the identity-based and morphing-based approaches, which refer to whether the chain of corresponding identities of classes across ontology versions is known or unknown, respectively. This aspect examines whether the identity of concepts is persistent (and known) across versions in time. Much philosophical debate is invested in how and by which properties one can identify a concept across time and how this can be formalized [20]. Some approaches utilize the notions of perdurance and endurance, as defined in [21], so as to seek identity. Specifically, looking at rigid properties that have to be persistent for all instances of a concept can be used to identify entities [22].

Further works have followed up, focusing on the extensional drift aspect of statistical data [14]. Given the high applicability of the method, we adopted it in this framework with the intention to extend and further define methods and tools for its wider adoption by the Semantic Web community.

The aforementioned methods to measure semantic drift share many common approaches with those within the field of ontology matching and alignment [23]. Ontology matching aims at finding correspondences between semantically related entities of different ontologies. These correspondences may stand for equivalence as well as other relations, such as consequence, subsumption,

or disjointness, between ontology entities. Ontology entities, in turn, usually denote the named entities of ontologies, such as classes, properties or individuals. However, these entities can also be more complex expressions, such as formulas, concept definitions, queries or term building expressions. Ontology matching results, called alignments, can thus express with various degrees of precision the relations between the ontologies under consideration [23]. The underlying matching techniques include name-based, structure-based and extension-based amongst others. Similar techniques are also found in the field of semantic Web Service matching and composition [24]. Many of those techniques are adopted and optimized, by e.g. selecting the proper string-matching algorithms, within the semantic drift metrics proposed in this work.

4. SemaDrift metrics and methods

This section presents our proposed framework for measuring semantic drift, i.e. a change in the meaning of a concept over time, location, culture, etc. Previous studies [13] have transferred highly applicable notions and metrics for measuring concept drift in the context of data mining to semantic drift. Specifically, the method to measure concept drift in semantics considers two basic pillars of change: (a) the different aspects of change, and (b) whether concept identity is known or not. The different types of change, reflecting its meaning, include:

- *Label*, which refers to the description of a concept, via its name or title;
- *Intension*, which refers to the characteristics implied by it, via its properties;
- *Extension*, which refers to the set of things it extends to, via its instances.

Meanwhile, the correspondence of a concept across versions can be either known or unknown, resulting in two known approaches for measuring change, along with a third, hybrid one proposed in this paper:

- *Identity-based approach* (i.e. known concept identity): Assessing the extent of shift or stability of a concept's meaning is performed under the assumption that its identity is known across ontologies. For instance, considering an ontology A, and its evolution, ontology B, each concept of A is known to correspond to a unique, known concept of B.
- *Morphing-based approach* (i.e. unknown concept identity): Each concept is pertaining to just a single moment in time (ontology), while its identity is unknown across versions (ontologies), as it constantly evolves/morphs into new, even highly similar, concepts. Therefore, its change has to be measured in comparison to every concept of an evolved ontology.
- *Hybrid approach* (i.e. best-match concept identity): Each concept's identity across versions (ontologies) is in fact unknown, similarly to the morphing-based approach. But after comparing a concept to all concepts of the following version, its identity is assumed to correspond to the most similar one. Therefore, it becomes known, and change is measured as in the identity-based approach.

Regarding the identity-based and hybrid approach, the assumption of single concept correspondence is made, in accordance to the notion of unique "identity", as initially introduced for the first approach in [13]. Notably, it does not prevent meaning to migrate to more than one concept independently from identity, which is after all, what semantic drift measures should trace.

The contribution of this paper in this area is the implementation and extension of these methods in the SemaDrift Application Suite, an open, reusable software solution, which was missing so far to the best of knowledge. Specifically, the methods proposed here concern the morphing-based and the hybrid approach, the only ones applicable in the general case of an out-of-the-box sequence of ontologies, without the need for human annotation of concept identity correspondence. Notably, the proposed morphing-based approach implementation was initially presented in [8], while the hybrid approach is entirely novel.

The following subsections describe the objective methods to compare concepts according to different aspects and the two methods to apply those metrics according to the morphing-based and the hybrid approach, respectively.

4.1. Description of aspects

This section considers the definition and description of objective aspects to generally capture a single concept's features and, consequently, compare it to another concept. This means that no specific approach of knowing the origin and identity is assumed. On the contrary, the identity, morphing-based and the hybrid approach introduced in this paper, which define the concepts to be compared using the aspects, are considered in the following sections.

Specifically, three different aspects are defined for capturing the meaning of each concept at a given point t (e.g. in time) as follows:

$$C^t = \langle label_t(C), int_t(C), ext_t(C)\rangle$$

where C^t denotes the meaning of concept C at point t, $label_t(C)$ denotes the label aspect of concept C at point t, $int_t(C)$ denotes the intensional aspect of concept C at point t and $ext_t(C)$ denotes the extensional aspect of concept C at point t.

Each aspect is resolved and formulated both intuitively and based on [13]. More specifically, label is selected as the most representative title description of a concept, intension refers to all properties that involve the concept, equally as subject or object, and the extension refers to the number of instances that belong to this concept. The aspects are defined as follows:

$$label_t(C) = \{l, \langle C, rdfs : label, l\rangle \in T\}$$
$$int_t(C) = \{i, i = \langle P, x, C\rangle \wedge (x = rdfs : domain \vee x = rdfs : range) \wedge i \in T\}$$
$$ext_t(C) = \{x, \langle x, rdf : type, C\rangle \in T\}$$

where T is the set of all triples in the ontology version t. Notably, no reasoning is performed prior to calculating the aspects in any of the proposed methods.

In other words:

- The label aspect is given by the $rdfs : label$ of a concept. The aspect assumes that each and every concept has exactly one label as in [7]. Any deviation from that, is regarded as a data quality issue that demands prior data cleaning e.g. to select one of multiple labels or fill in the ones missing.
- The intensional aspect is represented by the set of triples where C is either the $rdfs : domain$ or the $rdfs : range$ of a property.
- The extensional aspect is represented by the set of instances of class C (i.e. all instances that have class C as their $rdf : type$).

Overall, regarding data types, label is a string, intension is a set of triples of strings and extension is a set of strings. Notably, the resource identifiers within both intension triples of strings (properties) and extension strings (instances) are pruned. E.g. http://semanticweb.org/ontologies/mklab/2016/04/tate_2011.owl#hasSize becomes "hasSize". Removing the ontology names, which will always differ across versions, is necessary in order for

Table 2
Concept similarity, or else stability, metrics per aspect.

Aspect	Similarity/Stability metric
Label	String similarity with Monge–Elkan
Intension	Jaccard similarity between sets of triples
Extension	Jaccard similarity between sets of resources

the remaining string to completely match when comparing the sets. For this reason, we refer to pruned resource identifiers merely as "strings" and not as "resources". With pruning necessary, at least for the current approach, the methods assume that all URIs processed define fragments using a hash mark "#".

4.2. Description of methods

This section describes how different methods employ different similarity metrics to measure semantic drift in ontologies. This paper aims to provide tools and methods that remain generally applicable to ontologies out-of-the-box without significant human intervention and additional drift-related metadata. Although several methods have been proposed to seek identity correspondence across versions [22], they can be domain or model dependent, mandating for ad-hoc expert knowledge in the form of annotations, user input or using explicit identities. Therefore, between the two well-known approaches, identity-based and morphing-based [13], only the latter is considered. Additionally, an original, novel method, the hybrid approach, is introduced in this paper.

Both approaches use the same aspects as defined in the previous subsection. At its core, any approach has to measure differentiation between two concepts. Data types vary for each aspect and, hence, require different methods. The proposed similarity (or else stability) metrics chosen for each aspect are summarized in Table 2.

Specifically, to compare the strings acquired for the label aspect, the method adopted is the Monge–Elkan algorithm [25], which has been found to optimally suit string naming conventions frequently used in ontologies, such as CamelCase or snake_case, without the need for pre-processing (splits) [24]. Moreover, for both sets acquired for the intension and extension aspects, standard Jaccard similarity is used to compare them. Apart from the aforementioned metrics to compare concept aspects, the differences of the two methods are described below.

The following subsections describe the two different methods to measure semantic drift, according to the two respective approaches, the morphing-based and the hybrid one.

4.2.1. Morphing-based method

This approach assumes that concept identity across versions is unknown. Thus, each concept of an ontology version at point t_1 is compared to all concepts of the version next in order, at point t_2, for each of the aspects. Due to the morphing-based approach, all concepts at t_2 should account for measuring drift. Therefore, we consider the average similarity between C in t_1 and all concepts of t_2 so as to estimate drift between the two versions in all aspects.

For *label* drift, the two strings are compared based on the text similarity algorithm Monge–Elkan, for the aforementioned reasons. More precisely, in the morphing-based approach, the label drift of a concept C between versions t_1 and t_2 is defined as the average of Monge–Elkan text similarity between C in t_1 and all concepts of t_2:

$$label_{t_1\to t_2}(C) = \frac{\sum_{i=1}^{n_2} MongeElkan\left(label_{t_1}(C), label_{t_2}(C_i)\right)}{n_2}$$

where $label_{t_1\to t_2}(C)$ is the label drift of C between versions t_1 and t_2, $label_{t_1}(C)$ is a string representing the label aspect of C at point t_1 and n_2 is the total number of concepts in t_2.

In order to measure the similarity of two sets, we deploy the Jaccard similarity, which is defined as follows:

$$Jaccard(A, B) = \frac{A \cap B}{A \cup B}$$

where A, B are two sets of items. Based on that, we define the intensional drift of a concept C between versions t_1 and t_2 as the average of the Jaccard similarities between C in t_1 and all concepts of t_2. This is defined as:

$$int_{t_1\to t_2}(C) = \frac{\sum_{i=1}^{n_2} Jaccard\left(int_{t_1}(C), int_{t_2}(C_i)\right)}{n_2}$$

where $int_{t_1\to t_2}(C)$ is the intensional drift of C between versions t_1 and t_2, $int_{t_1}(C)$ is a set of triples representing the intension of C at point t_1 (properties) and n_2 is the total number of concepts in t_2.

Similarly, we define the extensional drift of concept C between versions t_1 and t_2 as the average of the Jaccard similarities between C in t_1 and all concepts of t_2.

$$ext_{t_1\to t_2}(C) = \frac{\sum_{i=1}^{n_2} Jaccard\left(ext_{t_1}(C), ext_{t_2}(C_i)\right)}{n_2}$$

where $ext_{t_1\to t_2}(C)$ is the extensional drift of C between versions t_1 and t_2, $ext_{t_1}(C)$ is a set of strings representing the extension of C at point t_1 (instances) and n_2 is the total number of concepts in t_2.

Finally, as all metrics have a range of [0, 1], their average (without using weights, in the general case) can be considered as an overall, "whole" aspect. Specifically, the whole drift of concept C between versions t_1 and t_2 is defined as the average of label, intensional and extensional drift between the two versions:

$$whole_{t_1\to t_2}(C) = \frac{label_{t_1\to t_2}(C) + int_{t_1\to t_2}(C) + ext_{t_1\to t_2}(C)}{3}$$

4.2.2. Hybrid method

The hybrid approach proposed in this paper combines aspects of both the identity- and the morphing-based approach to provide more conclusive insights. Initially, concept identity is in fact unknown, as no such metadata is guaranteed to be found attached to ontologies online, nor is the time and ability of domain experts to produce them. Therefore, as in the morphing-based approach, the hybrid algorithm measures concept similarity between a given concept and all concepts of the next version in line. However, instead of averaging the outcomes, it then assumes the identity of the concept by selecting the one with the highest similarity, which is, hence, equal to the concept stability metric across the two ontologies. As a result, the overall method remains generic enough, while it also provides the benefits of knowing concept identity, without the need for additional metadata.

In detail, in the hybrid approach the *label* drift of a concept C between versions t_1 and t_2 is defined as the maximum of Monge–Elkan text similarity between C in t_1 and all concepts of t_2:

$$label_{t_1\to t_2}(C) = \underset{i}{\operatorname{argmax}}\left(MongeElkan(label_{t_1}(C), label_{t_2}(C_i))\right), i \in [0, n_2]$$

where $label_{t_1\to t_2}(C)$ is the label drift of C between versions t_1 and t_2, $label_{t_1}(C)$ is a string representing the label aspect of C at point t_1 and n_2 is the total number of concepts in t_2.

Intensional drift is again based on Jaccard similarity. In the intensional drift of a concept C between versions t_1 and t_2 is defined as the maximum of the Jaccard similarity between C in t_1 and all concepts of t_2. This is defined as:

$$int_{t_1\to t_2}(C) = \underset{i}{\operatorname{argmax}}\left(Jaccard(int_{t_1}(C), int_{t_2}(C_i))\right), i \in [0, n_2]$$

where $int_{t_1\to t_2}(C)$ is the intensional drift of C between versions t_1 and t_2, $int_{t_1}(C)$ is a set of triples representing the intension of C at point t_1 (properties) and n_2 is the total number of concepts in t_2.

Likewise, the *extensional* drift of concept C between versions t_1 and t_2 is defined as the maximum Jaccard similarity between C in t_1 and all concepts of t_2.

$$ext_{t_1 \to t_2}(C) = argmax_i(Jaccard\left(ext_{t_1}(C), ext_{t_2}(C_i)\right)), i \in [0, n_2]$$

where $ext_{t_1 \to t_2}(C)$ is the extensional drift of C between versions t_1 and t_2, $ext_{t_1}(C)$ is a set of strings representing the extension of C at point t_1 (instances) and n_2 is the total number of concepts in t_2.

Notably, all the metrics always select exactly one concept of maximum similarity. If the maximum similarity value is shared between two or more concepts, the first concept in alphabetical order is always selected. No other tie-breaking mechanism has been explored yet. Also, as the formalisms imply, pairing is not exclusive either. Each concept identity is greedily assigned to the one with the maximum similarity heuristic — no search frontier for assigned concepts is kept. Therefore, one concept can function as the identity of more than one concepts in the previous ontology, without obstructing any of the methods. Future work can consider more optimal mechanisms to break ties, perhaps by assigning no more than one identity to each concept.

The *whole* drift of concept C between versions t_1 and t_2 remains the average of label, intensional and extensional drift between the two versions:

$$whole_{t_1 \to t_2}(C) = \frac{label_{t_1 \to t_2}(C) + int_{t_1 \to t_2}(C) + ext_{t_1 \to t_2}(C)}{3}$$

4.3. Chains and concept ranking

With the addition of the hybrid method, the notion of hybrid chains are introduced to the currently known morphing chains, along with the possibility to produce a new concept stability ranking. With this opportunity, the notions of a morphing chain is revised while hybrid chains and concept ranking are also described.

4.3.1. Morphing chains

One type of chains in the semantic drift domain is morphing chains, a term coined in [7]. They emerge from the morphing-based approach, which explores the similarity of a concept to all concepts of the consecutive ontology. Each chain begins from a certain root node of a chosen ontology. All concept nodes of the next ontology are linked to it with edges of different weights, equal to the respective similarity to the root concept. Each concept of the second ontology is linked to every concept of the next ontology and so on. In the end, the morphing chain is a weighted tree graph consisting of a single root branching out to every concept of the subsequent ontology, which in turn is linked to the next etc., as shown in Fig. 2. A different morphing chain can be constructed for any root concept and for each of the four aspects.

While the visualization of the morphing-chain may not always be compact, it is still a useful tool for experts to explore drift when following the morphing-based approach. A human observer may detect patterns and pinpoint the interesting, relatively large shifts of meaning across time through morphing chains. However, the amount of information can be overwhelming. Moreover, averaging similarity does not really provide useful drift information. It is more likely to provide the concepts which are similar to all concepts of the consecutive ontology. In other words, it can be a sign of homogeneity instead of stability.

4.3.2. Hybrid chains

A novelty of the hybrid approach are *chains*, which in turn allow finding *average concept chain stability* and *chain ranking* from the most stable to the least stable. In the identity-based approach, the exclusive concept correspondence is known (by humans). A concept corresponds to either one or none of the concepts of the next ontology version, as in same cases they may even disappear. Meanwhile, the hybrid approach, having no reliable method of knowing this, considers exactly one corresponding concept in each ontology version. Therefore, instead of a morphing chain, a *chain* can be constructed, linking each concept-node with only a single other concept-node, i.e. the one with the maximum stability. To obtain this, the SemaDrift algorithm begins with the first ontology in the chain and links each of its concepts to a single concept of the consecutive ontology. Unlinked nodes, i.e. nodes that were not matched as corresponding ones of the former, become the head nodes initiating new chains. Then, the nodes of the second ontology are linked to the third and so on.

An example generation of hybrid chains is shown on Fig. 3. Starting from the first ontology's starting node, the most similar node of the next ontology is linked. In this example, two unlinked nodes emerge in consecutive versions, each starting a new chain. Visualizing the chains can provide experts with a comprehensive outlook of concept correspondence across multiple versions along with the level of similarity between each of them. Both morphing chains and chains, are explored and exploited in the SemaDrift Application Suite as described in the following sections.

4.3.3. Concept ranking

An added advantage of hybrid chains is that average concept chain stability can now be calculated by averaging all concept stability values across each chain. That can, in turn, provide a ranking of the chains from higher to lower average stability. Examining the head, i.e. the root concept of each chain, we can indirectly deduce the stability of concepts themselves across the ontology chain; an interesting insight which was unavailable prior to the hybrid method. In the example of Fig. 3, the algorithm will return the chain heads along with the average stability from higher to lower as follows:

- *tate*_2011.*owl#SoftwareBased*: 0.9165
- *tate*_2007.*owl#MixedMedia*: 0.9046
- *tate*_2003.*owl#ComputerBased*: 0.89575

This aids the scholar to immediately observe the most and the least stable chains (and, in a sense, concepts), a useful feature of the identity-based method [7], that now becomes available in the hybrid approach.

5. SemaDrift application suite

The SemaDrift Applications Suite is a collection of software tools to aid measuring semantic drift in ontologies. The tools are intended to cover a wide range of end-users and usage scenarios, from developers and knowledge engineers to domain experts such as archaeologists, biologists, artists, managers and endless additional professions. All the components of the suite are currently available online[2] under Apache V2 license, to encourage re-use, reproducing results and community contributions.

To begin with, the SemaDrift Library is essentially a software library implementing the aforementioned methods and utilities to load ontologies and calculate the drift metrics. This API is intended for developers, but also serves as a basis to develop end-user applications for experts and non-experts alike. The two applications, SemaDrift Protégé Plugin and SemaDriftFx, serve as two such examples, but also as standalone tools. The former is intended for ontology engineers using the Protégé ontology authoring environment [26], while the latter provides more advanced utilities and visual output.

Each component of the suit is presented in its own subsection, while the last subsection shows a comparative overview of all of them, a complete architecture, their interrelationships and complementary roles.

[2] http://mklab.iti.gr/project/semadrift.

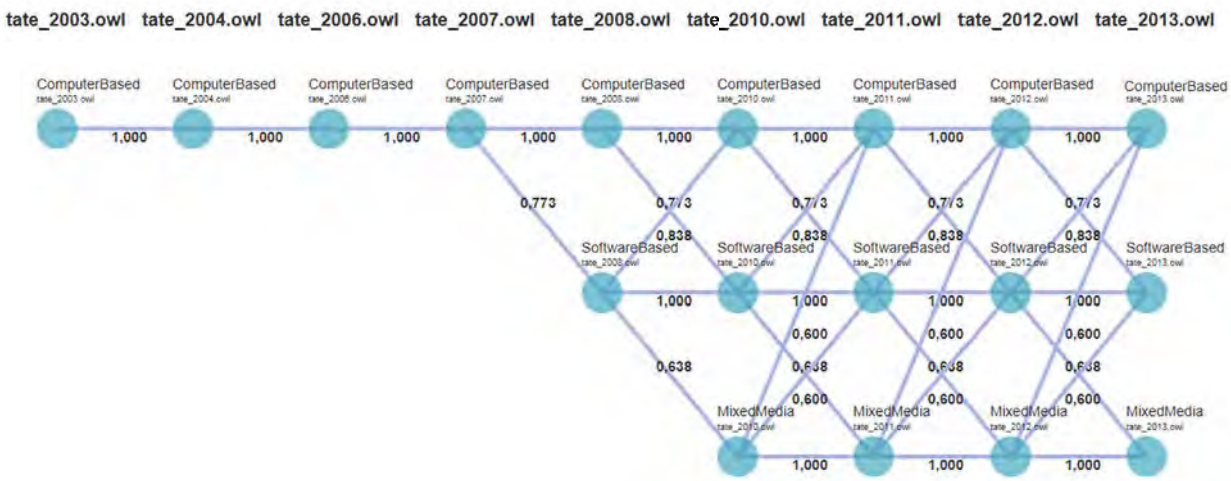


Fig. 2. An example of a morphing chain. From the starting node, all nodes of the next ontologies are linked with corresponding similarity measures.

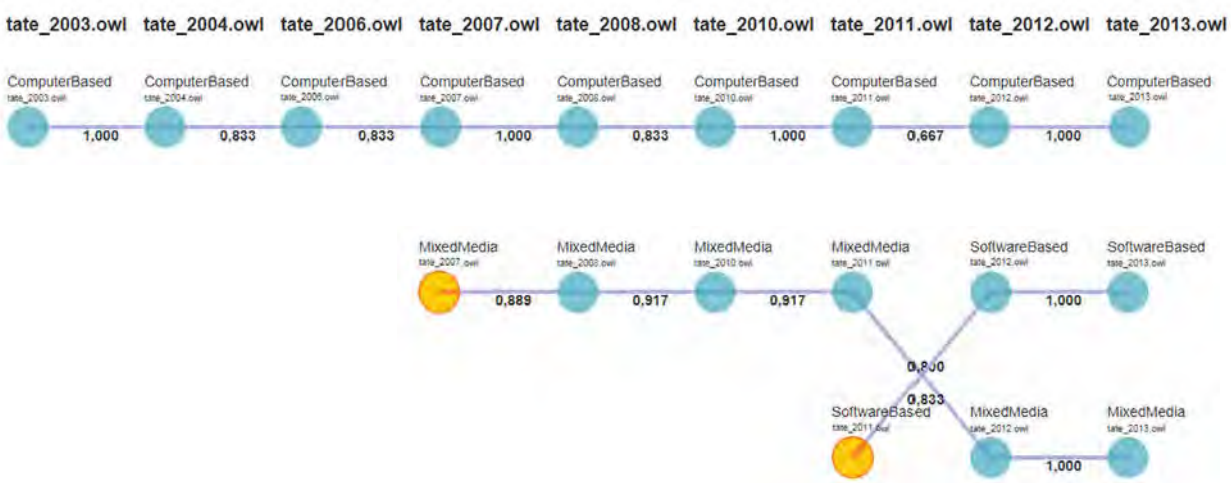


Fig. 3. An example of a hybrid chain. Only the nodes of maximum similarity appear at each step. Unlinked nodes, marked in yellow, appear as head nodes starting new chains.

5.1. SemaDrift library API

The methods described in Section 4 were implemented as a software library, in order to allow reuse, encourage research in the field of semantic drift measurement and reproducing research results in an objective manner. The platform-independent library is written in Java and provides an API for programmatic and/or command-line usage. The OWL-API library[3] was deployed for handling RDF/OWL operations, while the Simmetrics library[4] provided the implementation of Monge–Elkan text similarity measure.

Specifically, the SemaDrift Library API features the following core functions, along with some utilities and abstractions, useful for building higher level applications.

5.1.1. Managing the ontology chain

With regards to managing the ontology chain, the following operations are allowed:

- Add a new ontology in the chain using a local file path, URL or "OWLOntology" object (OWL-API);
- Add a set of ontologies in the chain using a set of file paths or URLs;
- Remove an ontology from a specific position in the chain;
- Empty chain, i.e. remove all ontologies from the chain;
- Get all ontologies in the chain.

Notably, for managing the order of the ontologies, which of course determines results, users need to remove and add ontologies accordingly in a stack-like manner, or build their own utilities to do so.

5.1.2. Accessing the ontology

Regarding access to the ontology, the following list of operations are provided:

- Load an ontology from a local file path URL as an API object;
- Get an ontology's set of classes;
- Get ontology tree structure, i.e. classes linked to their subclasses;
- Get ontology class name, IRI, subclasses and properties;
- Get ontology property name, domain and range.

5.1.3. Metrics for all aspects

Regarding the morphing-based approach, the following metrics are featured:

- Get *Concept-per-Concept Stability*, namely the stability between all concept pairs of each pair of consecutive

[3] http://owlapi.sourceforge.net/.
[4] https://github.com/Simmetrics/simmetrics.

ontologies in the chain, i.e. for each concept of the former to all concepts of the latter ontology. Since this metric links all concepts of an ontology to all concepts of the consecutive one, it can also be seen as a *Morphing Chain*.
- Get the *Average Concept Stability*, i.e. for each pair of consecutive ontologies, get the average concept stability of all concepts of the former ontology; this is the *average* of the stability between the given concept and all the concepts of the latter ontology.

On the other hand, with regards to the hybrid approach, the following metrics are provided by the API:

- *Maximum Concept-per-Concept Stability* values, which means the maximum stability between each concept of the former ontology and all the concepts of the latter ontology in each ontology pair. Since this metric uniquely links all concepts to just one consecutive ontology concept, it can also be seen as a *Chain*.
- Get the *Average Concept Stability*, i.e. for each pair of consecutive ontologies, get the average concept stability of all concepts of the former ontology; which is the *maximum* of the stability between the given concept and all the concepts of the latter ontology.
- *Concept Chains*, i.e. construct chains by linking each concept of the former ontology with the concept of maximum stability in the latter ontology, for all ontology pairs.
- *Concept Chain Ranking*, i.e. get and sort the average concept stability values across each chain.
- *Concept Ranking* by getting the head concept of each chain in the Chain Ranking from the most to the least stable.

5.2. SemaDrift Protégé Plugin

This component of the SemaDrift Application Suite aims to bring novel semantic drift measuring capabilities into a popular ontology authoring platform, Protégé. Through this, end-users can have reviewing and editing capabilities on an ontology while simultaneously viewing drift metrics in the same environment.

Protégé offers many advantages to be chosen as the tool to integrate with. Traditionally as a desktop application, and recently also as a web application, it provides a user-friendly graphical interface for authoring ontologies, and naturally constitutes a more flexible alternative to plain text or RDF/OWL, especially for the unfamiliarized users. In detail, its main advantages for plug-ins are:

1. Large user community: Protégé currently counts more than 320.000 users.[5]
2. Robust development environment, open source, active mailing list and wiki.
3. Multi-purpose plugins enabling querying, visualization and direct reasoning in the same environment.
4. Reuse of powerful authoring, reasoning and visualization tools for knowledge engineers: Each plugin is able to reuse the existing infrastructure. In the case of semantic drift measurement, it provides appealing ontology visualization techniques in order to examine causes of measured drift.

5.2.1. Functionality

An overall look at the SemaDrift Protégé Plugin functionality is shown in Fig. 4. The tool provides a subset of the basic functions of the underlying SemaDrift API, in a graphical environment. For that purpose, it exposes some of its functions and accommodates the outcomes in suitable user controls using the Java Swing library. This edition of the plugin focuses on ontology pairs, i.e. two versions of the same ontology, in order to provide more insight into their differences, fitting also into the Protégé workspace philosophy. Usually, the users work on a single ontology at a time, which is always displayed as a tree hierarchy of classes in the left pane. Then, plugins occupy the right pane, which is free to accommodate their functions (Fig. 4). Naturally, the tables on the plugin panel shows stability metrics for every concept in the ontology, but for the sake of the paper only three of them are displayed here.

As a first step the user has to select the pair of ontologies for which to measure drift. To take advantage of the environment, the plugin assumes that the first selected ontology is the one currently loaded in Protégé, allowing also its in-depth visualization, reasoning and query execution. The second ontology can be selected from the SemaDrift pane using the "Browse" button to look through local or remote storage.

After both ontologies are available, pressing on the "Measure Drift" button will display the SemaDrift metric results. Stability, as a measure of drift, is shown in two sections: overall average stability per aspect and concept pair stability for all aspects. The first section constitutes the most generic, abstract measure of drift. It displays a table with the average drift of all concepts from the former ontology to the latter, per each of the four aspects: label, intension, extension and whole. Naturally, the measurements are derived using the metrics and algorithms for each aspect described in the previous section, yielding a value from zero (no similarity) to one (full similarity).

The second section of results is displayed in respective tables. Each table row corresponds to a concept of the former ontology and each column to a concept of the latter. Consequently, each cell holds the similarity metric (i.e. concept stability) between each pair of concepts. These similarity values between pairs can further be utilized by users for different purposes; examples are given in the next section.

Concluding, the plugin in its current form is in essence a first step towards measuring semantic drift in a graphical manner. Some of its many possible extensions are implemented in the SemaDriftFx application introduced in the next section, while others are given in the final section of this paper.

5.3. SemaDriftFx application

In contrast to the Protégé Plugin component, SemaDriftFx is a standalone application. This way it complements the Application Suite by offering more flexibility, more graphical utilities, and focused usage, capitalizing on all the underlying methods and metrics offered by the SemaDrift Library API. While it still offers some ontology exploring capabilities, it omits the ability to author and edit them, functions that would overwhelm domain experts.

5.3.1. Main window — ontology chain

Fig. 6 illustrates the main window of SemaDriftFx. Its main content is the ontology workspace, where users are able to create, explore and rearrange an ontology chain (i.e. a series of ontology versions) in an array of ontology containers. Additional buttons in the toolbar are used to calculate drift measurements, visualize chains and display useful information.

Initially, the only available action is to start creating the ontology chain. Starting from a single, empty container, the user can add entries to the chain by using the "+" button at the end of the chain, which will add a new empty container. Every container can be managed by its own toolbar at the top, offering the ability to load an ontology, reload it in case it has been edited externally, and remove it from the chain. Additionally the left and right arrows

[5] https://protege.stanford.edu/.

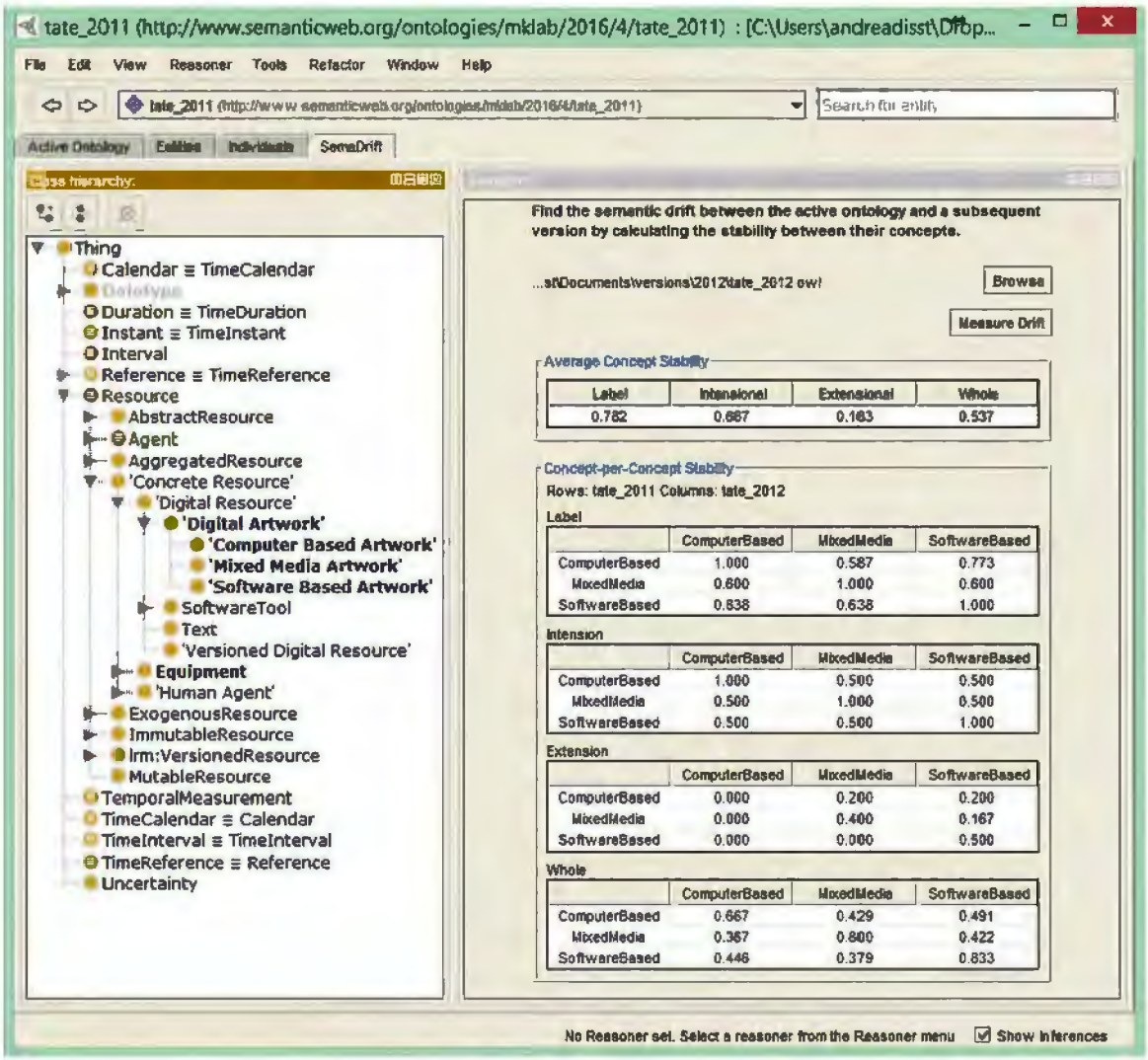


Fig. 4. The SemaDrift Protégé Plugin main window shows the native tree hierarchy of the ontology currently open (left), with the option to select a second ontology and view drift metrics (right).

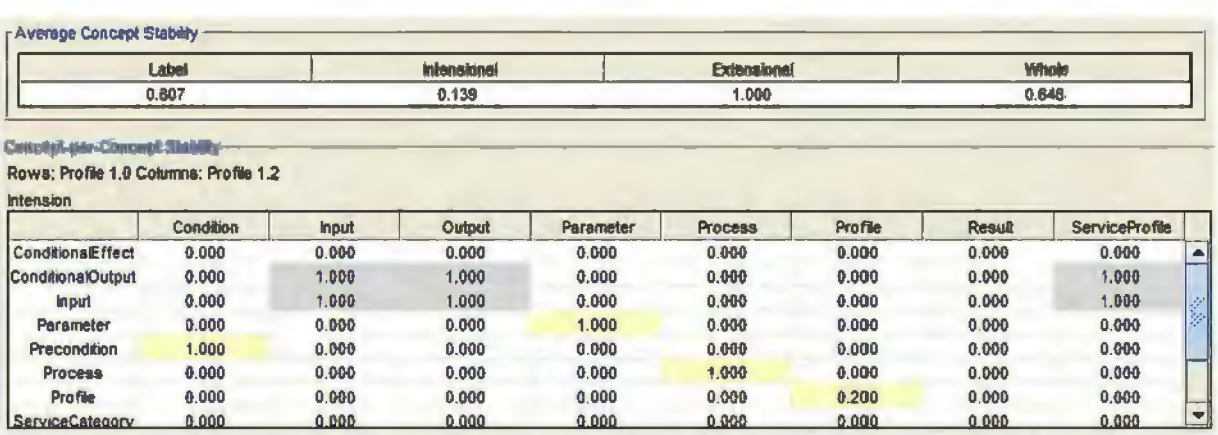


Fig. 5. SemaDrift Protégé Plugin showing average concept and concept-per-concept intension stability for OWL-S ontology for semantic markup of versions 1.0 and 1.2.

move it up and down the chain, respectively. It is up to the user to order the ontology versions, usually chronologically, so as to produce the correct drift measurements. To identify and explore the ontologies while constructing the chain, each container displays the hierarchy tree of ontology classes and properties, while the ontology filename is displayed at the bottom. Object and datatype properties are displayed in blue and green icons respectively, easily distinguishable also by their displayed range values.

Fig. 6. The SemaDriftFx application main window, allowing users to create an array of ontologies, view tree hierarchies (bottom), calculate drift metrics and show visualizations (top).

5.3.2. Measuring drift

After loading at least two ontologies in the chain, the "Measure Drift" action button becomes available at the toolbar. The action executes drift measurement methods in the background and displays a pop-up window with the output, as shown on Fig. 7. Quite similarly to the Protégé plugin, the window displays average concept stability per aspect, but this time for the entire ontology series. Next, concept-per-concept stability metrics are displayed in a matrix. A matrix is displayed for each aspect and ontology pair, organized in different numbered pages. Each page displays four matrices, one for each aspect, for a single ontology pair (the first page displays metrics for the first and second ontology, the second for the second and third, and so on). Users can switch pages by clicking the page number (in a tab-like fashion), while the ontology names are displayed to aid the user.

The results can be utilized in a number of ways. First of all, some insights might already be visible to the eye, especially for average per aspect stability. For the detailed concept stability, interesting shifts may still be highlighted, especially when few values are showing change, in relatively small ontologies. For larger ontologies, the metric values can be exported for use in external programs. For this reason the application offers an export to universal, CSV format function on the upper right corner of the Measure Drift window (Fig. 7). Most importantly, they are utilized by the application itself to visualize chains, as shown next.

5.3.3. Visualize chains

When drift is already measured for the present ontology chain in the main window, the "Visualize Chain" action button becomes available. It displays another pop-up window where visual graphs representing chains can be visualized (Fig. 8). There are two kinds of graphs displayed; either morphing-chains, following a one-to-many principle, or hybrid chains, following a one-to-one principle, as the underlying morphing-based and hybrid approaches mandate. In both cases, the graph behaves similarly.

The filtering options from the top toolbar allow selecting a "From" and a "To" ontology from respective drop-down boxes. Then, the chain will include all entries from the former to the latter ontology. The next dropdown box is used to select one of the available classes of the "From" ontology, to serve as the graph's root, i.e. the root concept to investigate. Another dropdown is used to select the aspect for which drift will be considered for the chain. As chains are dynamically visualized in the main area of the window, the user can enter a threshold in the respective numeric textbox and press the "Refresh" button. This action filters out concepts (vertices and nodes) that are linked to the graph by stability values lower than the threshold and clear out the chain to shifts that matter.

The main window displays graphs which can be manipulated in layers from left to right, in accordance to the ontology series. The manipulation mode decides the action of clicking and dragging within the graph. The active manipulation mode is displayed in the bottom left corner and can be one of the following:

1. Pan & Zoom: Click and drag pans the graph, mouse wheel zooms;
2. Clicking a node selects or deselects it. Dragging moves the selected group of nodes.

Although the chains are already arranged in a meaningful manner, the application allows rearranging nodes or groups of nodes, panning and zooming. Fig. 9 shows a rearranged chain. Changes of interest can be highlighted by selecting nodes (displayed in yellow) and hovering the mouse over a pair (displayed in bold). When the user is satisfied with the result, the "Save" button can be used to export the current instance of the graph as an image for reuse outside the application.

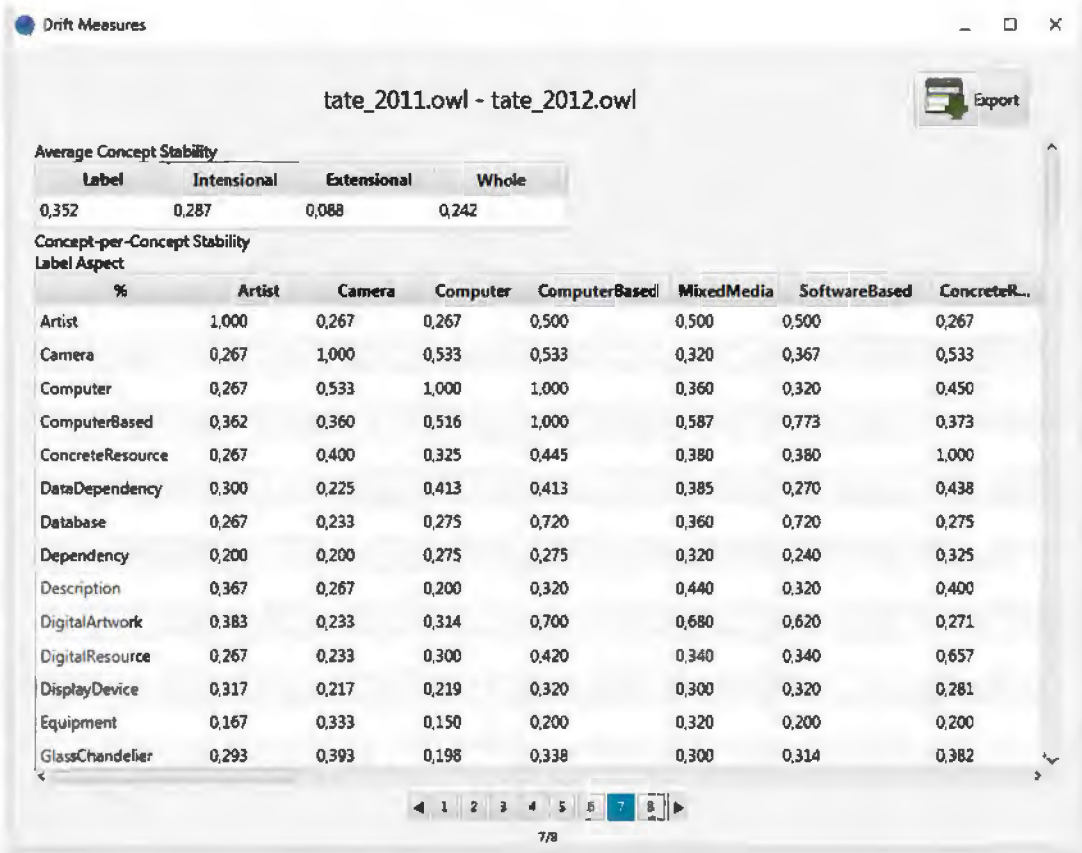


Fig. 7. The SemaDriftFx Measure Drift pop-up window, showing overall Average Concept Stability and Concept per Concept stability for all aspects and ontology pairs (in numbered pages), allowing also to export outcomes in CSV format.

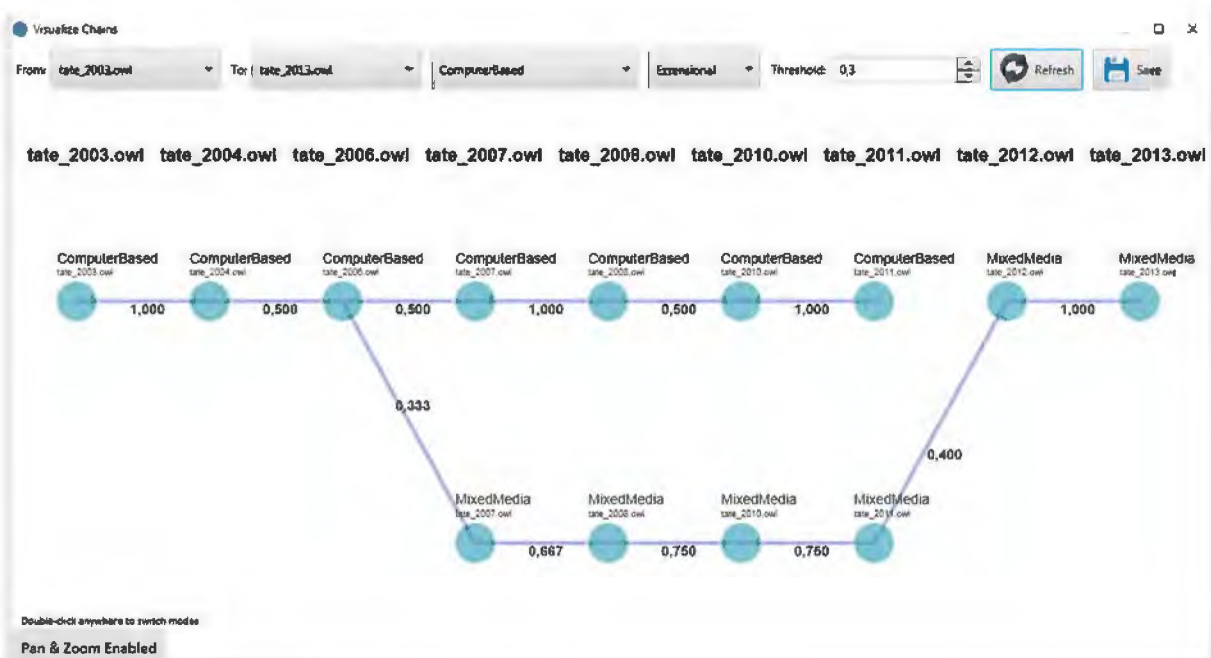


Fig. 8. The SemaDriftFx Visualize Chains pop-up window showing a morphing chain of the extensional aspect, with 0.3 stability threshold.

5.4. Application suite overview

A concept architecture of all applications in the suite so far, is depicted in Fig. 10. The three main applications are shown in white: SemaDrift Protégé Plugin, SemaDrift Library and SemaDriftFx, while software libraries are shown in blue. Their relationships and user interactions are also drawn. A core ontology parser is used for reusability and future compatibility, in our case OWL-API [27],

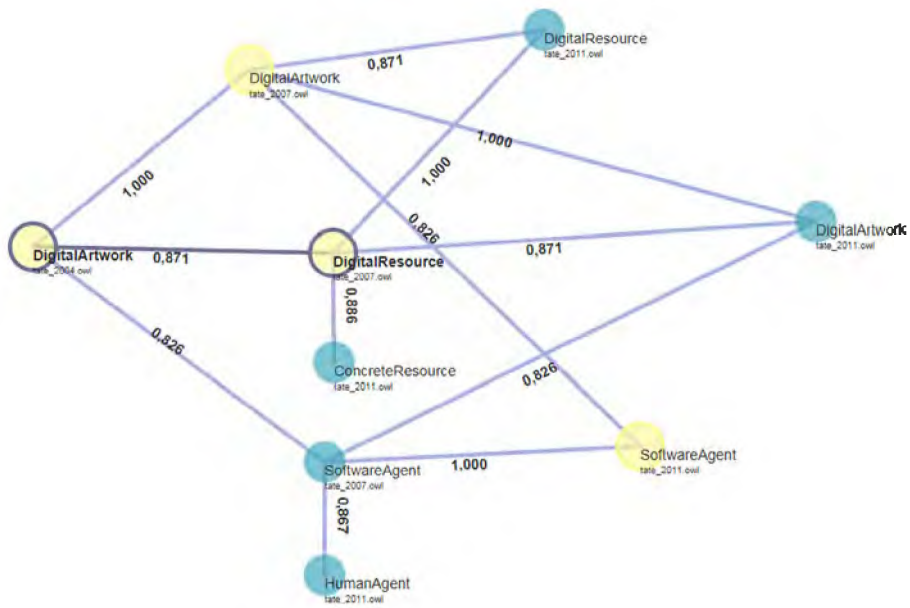


Fig. 9. An image extracted from SemaDriftFx with the morphing chain arranged in the desired manner, selected nodes in yellow and a mouse hovered node pair in bold. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

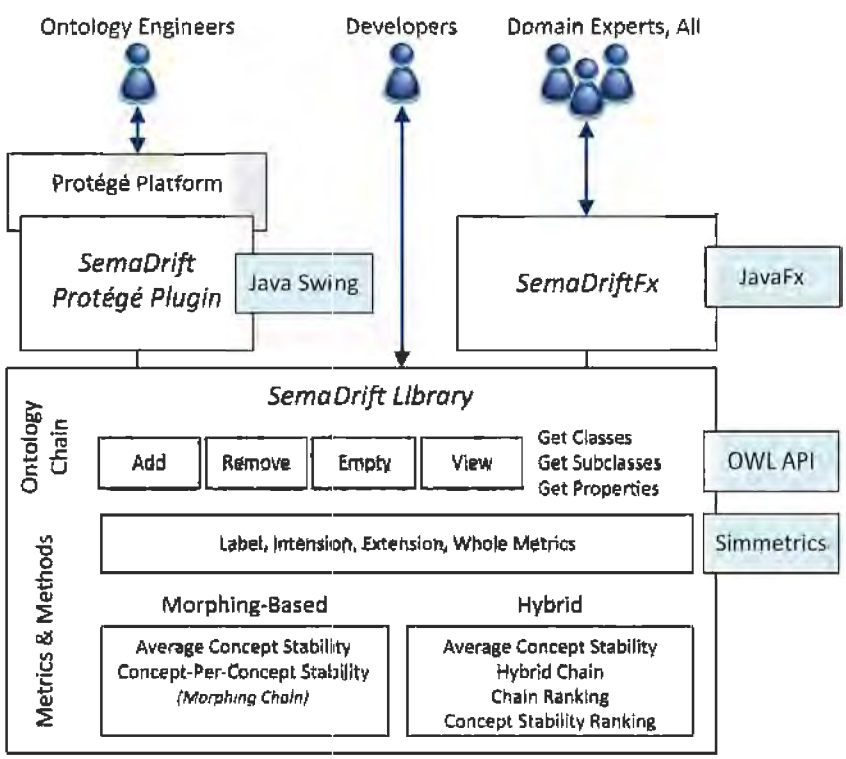


Fig. 10. A conceptual architecture of the SemaDrift Application Suite, showing components, software libraries used and user interactions.

while the Simmetrics library provides the Java implementation of text-similarity metrics. The Protégé plugin is integrated with the Protégé platform and developed with a compatible Java graphics and user controls library, Java Swing. SemaDriftFx is a standalone application exposing and exploiting more of the internal SemaDrift functions due to the usage of a more advanced framework, JavaFx. The library API is exposed itself for developers to create their own applications and experiments. All tools are developed in Java and provided as open-source to encourage developer reuse, extensions and contributions.

Table 3 displays a list of all functionality allowed in each tool of the suite, and its intended audience. These attributes are further justified in the following sections.

6. Use case scenarios

6.1. Digital preservation of art

Digital Preservation is an interdisciplinary field where long-term change insights are crucial, involving experts in arts, culture,

Table 3
Features of each component of the SemaDrift Application Suite.

Feature	SemaDrift library	SemaDrift Protégé Plugin	SemaDriftFx
Target users	Developers	Ontology Engineers	Domain experts
GUI	–	✓	✓
Load Ontologies	Unlimited	2	Unlimited
Re-arrange Ontologies	✓	–	✓
Edit Ontologies	–	1st Ontology	-
View Tree Hierarchy	All Ontologies	1st Ontology	All Ontologies
Average Concept Stability	✓	✓	✓
Concept-Per-Concept Stability	✓	✓	✓
Morphing-chains	✓	–	✓ Graph
Hybrid chains	✓	–	✓ Graph
Export image, CSV	–	-	✓

history and literature [28]. We applied the methods of the SemaDrift framework on an appropriate dataset, yielding interesting insights regarding semantic drift across time, which would otherwise be inaccessible.

6.1.1. Dataset

As the underlying knowledge representation structure, we extended a domain ontology representing Software-Based Art (SBA) [29–31], which was developed within the PERICLES EU FP7 project.[6] Based on the acquisition log by PERICLES partner Tate Galleries, London, describing the changes to Tate's cataloging of software-based artworks in the period 2003–2013, we populated the ontology with instances of the artworks and created respective consecutive versions of the ontology. During the period 2003–2013, three terms were mainly used for indexing the artworks: "Computer-based Art", "Mixed-Media Art" and "Software-based Art". Thus, a key problem to address was to investigate whether the three terms refer to semantically similar or different notions. More specifically, we wanted to measure the evolution and drift of the three relevant concepts: Computer-based (CB), Mixed-Media (MM) and Software-based (SB) art.

Fig. 12(a) illustrates the classification of the eight SBA artworks (left-most column) in the three concepts per year (rest of columns). Thus, the dataset contains a total of nine semantic models (ontologies) for this period, one model per year, excluding the years when no changes in the cataloging occurred.

6.1.2. Morphing-based approach

We fed the SBA ontology versions, ordered by year, to SemaDrift and started measuring the semantic drift. The output is presented here, starting from numeric measures, morphing chains and then an overall graph showing the different measures of stability across versions. Concept stability is measured as similarity, in the range of 0 (completely disjoint) to 1 (entirely identical label/ properties/ instances), according to label, intensional and extensional drift, respectively.

Initially, the metrics themselves show low stability in the extensional aspect in average concept stability 7. Scrolling down to the extensional aspect, indeed, in years 2011 to 2012, drift can be suspected from CB to MM and SB, while the CB concept itself disappears (zero similarity): Fig. 11. However, such information is hard to pinpoint with the naked eye within the matrix, without filtering or processing it externally. Therefore, moving on to morphing chains, our aim is to validate how easily this information will be apparent there.

Morphing chains show in detail concept similarity for each aspect, and how concept meaning migrates from one concept to another each year, from 2003 to 2013. Inspecting the label aspect shown in Fig. 12(b), shows concepts having similar names (values higher than 0.6), but mostly homonymous concepts remain in the ontology across versions (1.0 values) the highest similarity measure holds between concepts with the same name across versions, demonstrating stability. Likewise, the intensional aspect, shown in Fig. 12(c), demonstrates equal stability, as properties do not vary significantly across versions. Interestingly enough, all similar concepts are identical as they share the same properties (1.0 stability). Therefore, no real migration of properties from one concept to another occurs.

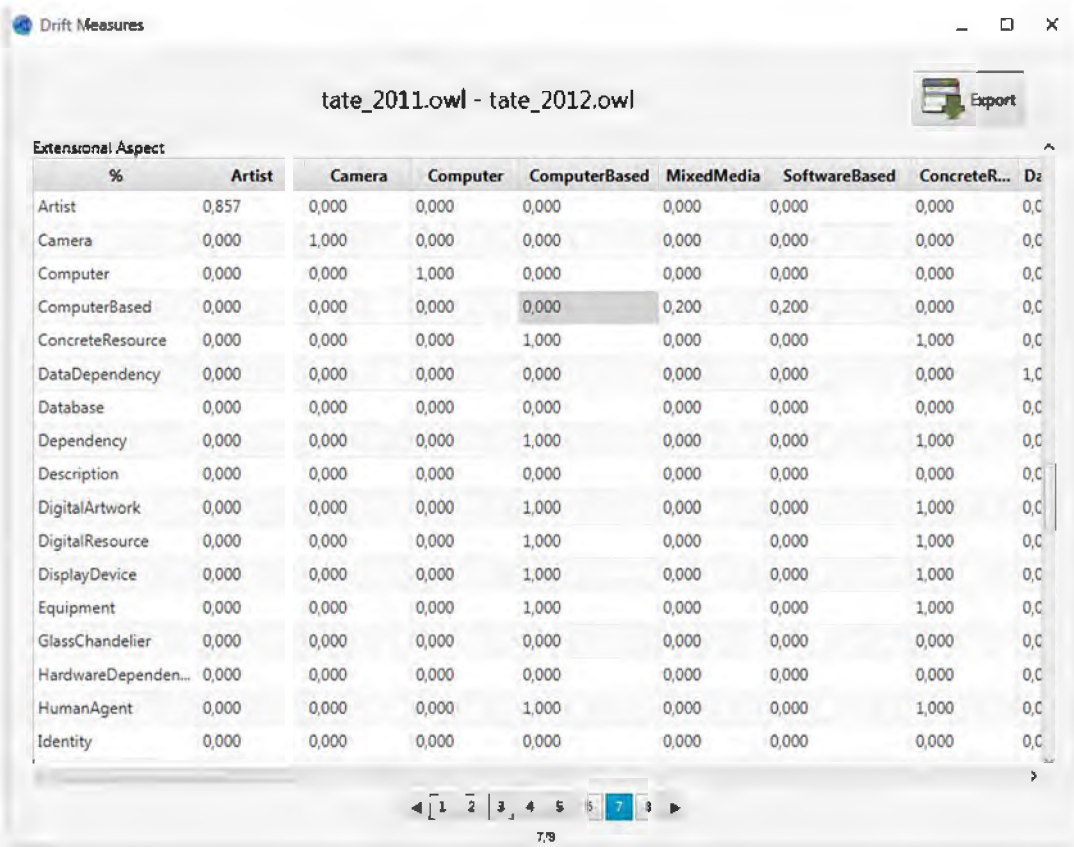
On the contrary, the extensional aspect, shown in Fig. 12(d), demonstrates variations from version to version, with the most significant one being the complete migration of the CB concept between 2011 and 2012, partially to MM and to SB, due to its instances shifting type. This development is validated by the experts as it coincided with an actual Tate policy introduced in 2011, according to which the "Computer-based" term was officially abandoned as not sufficiently descriptive and would no longer be used for characterizing the artworks. This demonstrates the capability of the developed methods to validate drift observed by domain experts. Another capability of the method is to detect splits and merges, which normally have to do most with the extensional aspect. Indeed, in this aspect, CB splits to two parts, one part merges with MM and the other merges with SB.

Finally, as seen in Fig. 12(e), the whole aspect depicts these changes in the greater scale, reflecting stabilities (due to label and intension) and some instabilities (due to extension). With the ability of the tool to export the metrics in CSV and process them, further graphs can be generated on-demand. As an example, averaging each aspect for all concepts per version reveals concept stability per aspect over time as shown in Fig. 12(f). This revealing graphic representation clearly shows at a glance that:

- The label is the most stable aspect, followed by intension, since labels and properties remain quite constant in the sample dataset.
- The extensional aspect is the least stable, as all instances of Computer-Based type are eventually evolved into Mixed-Media or Software-Based, as already discussed.
- Stability is reduced in all aspects during 2003–2008, as the ontology is enriched with new concepts.
- After looking at concept stability across versions, the Computer-Based concept appears to be the most stable one in all aspects. Notably, we cannot technically average stability for a concept across versions, since in the morphing-based approach their correspondence is unknown. However, we might still measure stability at each given pair, where the Computer-Based concept always ranks first. Still, examining the model more closely reveals that this could actually be attributed to its high similarity to the other two concepts and not to stability per se.

As previously discussed, the benefit of the morphing-based approach is that it clearly shows migrating meaning from one concept to multiple others. To do so, and due to unknown concept correspondence, the stability of a concept is measured across all concepts of the ontology version that follows. The pitfall here is that high similarity of a single concept with many concepts of the following ontology appears as high stability, while, oftentimes, it is in fact only circumstantial likeness to most ontology concepts. In these cases, this can be better interpreted as homogeneity instead of stability. This limitation of the morphing-based approach could be lifted by enriching the ontology with more concepts that will naturally vary more. With the identity-based approach not always accessible, the hybrid approach, presented in the next subsection, seems to be a promising alternative.

[6] http://pericles-project.eu/.



%	Artist	Camera	Computer	ComputerBased	MixedMedia	SoftwareBased	ConcreteR...	Da
Artist	0,857	0,000	0,000	0,000	0,000	0,000	0,000	0,0
Camera	0,000	1,000	0,000	0,000	0,000	0,000	0,000	0,0
Computer	0,000	0,000	1,000	0,000	0,000	0,000	0,000	0,0
ComputerBased	0,000	0,000	0,000	0,000	0,200	0,200	0,000	0,0
ConcreteResource	0,000	0,000	0,000	1,000	0,000	0,000	1,000	0,0
DataDependency	0,000	0,000	0,000	0,000	0,000	0,000	0,000	1,0
Database	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,0
Dependency	0,000	0,000	0,000	1,000	0,000	0,000	1,000	0,0
Description	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,0
DigitalArtwork	0,000	0,000	0,000	1,000	0,000	0,000	1,000	0,0
DigitalResource	0,000	0,000	0,000	1,000	0,000	0,000	1,000	0,0
DisplayDevice	0,000	0,000	0,000	1,000	0,000	0,000	1,000	0,0
Equipment	0,000	0,000	0,000	1,000	0,000	0,000	1,000	0,0
GlassChandelier	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,0
HardwareDependen...	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,0
HumanAgent	0,000	0,000	0,000	1,000	0,000	0,000	1,000	0,0
Identity	0,000	0,000	0,000	0,000	0,000	0,000	0,000	0,0

Fig. 11. Extensional aspect metrics show shift of the ComputerBased concept towards the MixedMedia and SoftwareBased concepts instead of the homonymous one.

6.1.3. Hybrid approach

The hybrid approach helps assume one corresponding identity for each concept, simplifying the process and reducing the amount of information, that might be overwhelming for the user. In this case, the label and intensional aspect do not present much interest, as in both cases the homonymous concepts are linked (due to all class stability being 1.0). Therefore, We examine the extensional aspect as shown on Fig. 13. While the new concepts MM and SB remain unchained, as far as the hybrid method is concerned, the CB concept clearly transforms to MM. The whole aspect also appears interesting, as it combines the extensional aspect, which is the one actually changing, label, a slightly changing aspect and intensional, a constant, unanimous aspect. As shown on Fig. 14, while stability fluctuates, the concepts remain mostly linked to the homonymous ones. Although these insights might not be as groundbreaking for the ontology at hand, it is apparent how the method can simplify and prune a larger graph.

More importantly, the hybrid approach allows for more easy-to-use metrics such as a simple concept stability ranking over the entire ontology series. While, admittedly in this use case, the chains are not dramatically different between the morphing and the hybrid approach, the ranking makes a lot of sense. According to both the extensional (Table 4) and the whole aspect (Table 5) the newest concept, SB, is the most stable one, while the oldest, CB, has undergone the most change.

6.2. Semantic markup of web services

The second scenario focuses on the OWL-S ontology for semantic markup of Web Services,[7] also known as Semantic Web Services.

[7] https://www.w3.org/Submission/OWL-S/.

Table 4
Ranking of digital preservation concepts according to hybrid chain average stability, extensional aspect.

Ranking	Concept	Hybrid chain stability
1	SoftwareBased	0.75
2	MixedMedia	0.7134
3	ComputerBased	0.7125

Table 5
Ranking of digital preservation concepts according to hybrid chain average stability, whole aspect.

Ranking	Concept	Hybrid Chain Stability
1	SoftwareBased	0.9165
2	MixedMedia	0.9046
3	ComputerBased	0.8957

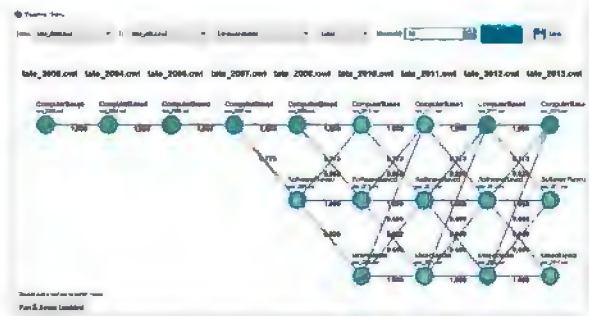
6.2.1. Dataset

Versions 1.0 and 1.2 of OWL-S available online were obtained. These versions are also accompanied with a thorough definition that will be used as "ground-truth" to verify any observations made via the tools. In OWL-S, each service has a Profile, a Grounding and a Process Model. A critical piece of Profile metadata are operation IOPEs, defining its Input and Output information (e.g. credit card number and total price), Preconditions model the requirements to invoke it (e.g. credit card clearance) and its Effects model the resulting outcomes (e.g. transferring ownership of goods or granting access).

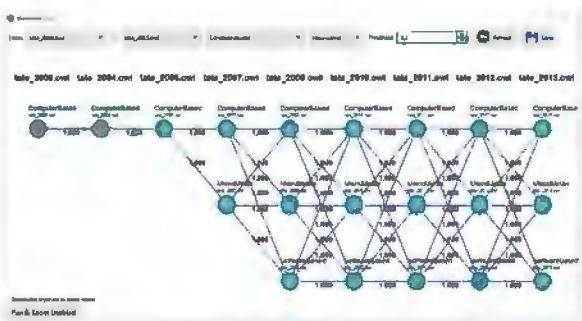
The aim of this scenario is to use a standardized ontology, commonly used and found online along with two of its versions and investigate how well the tools and methods can pinpoint semantic drift between them. No preprocessing will be made to the ontologies, while outcomes can be verified with the official explanation of the standards changing between versions.

	2003	2004	2006	2007	2008	2010	2011	2012	2013
T11812	CB	CB	CB	CB	CB	CB	CB	SB	SB
T12656			CB	MM	MM	MM	MM	SB	SB
T13251				MM	MM	MM	SB	SB	SB
X26607					MM	MM	MM	MM	MM
T13348					SB	SB	SB	SB	SB
T13645						CB	CB	MM	MM
L02995						MM	MM	MM	MM
X43413								MM	MM

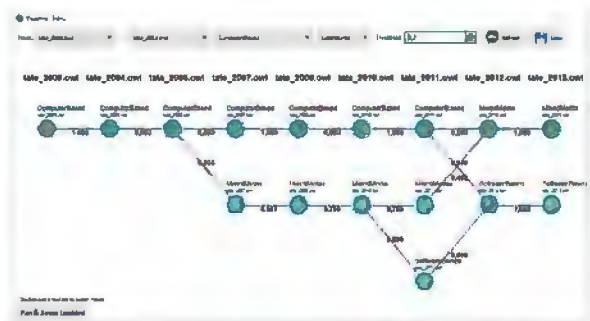
(a) Classification of SBA instances to the three concepts in the ontology. The left-most column contains the cataloging IDs of the artworks.



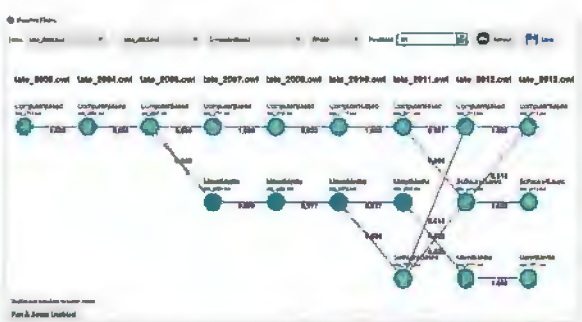
(b) Label aspect.



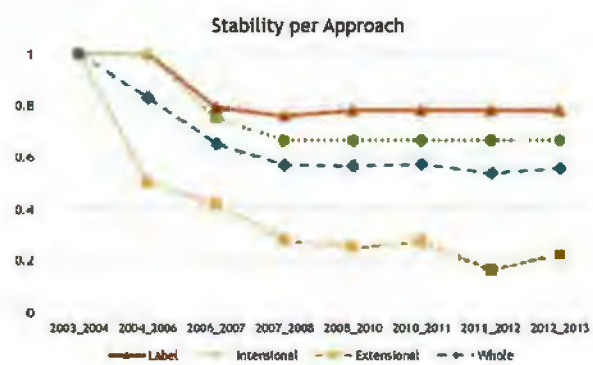
(c) Intensional aspect.



(d) Extensional aspect.



(e) Whole aspect.



(f) Average concept stability per aspect, across time.

Fig. 12. Morphing-based approach insights in the Digital Preservation use case scenario.

6.2.2. *Morphing-based approach*

In this scenario, the Profile ontology changes are immediately apparent, even in the numeric output of the methods. Fig. 5 shows average concept stability per aspect and concept-per-concept stability for the intension aspect (using the SemaDrift Protégé Plugin).

The greatest drift seems to occur within the intensional aspect, which leads us to investigate it further. Meanwhile, no instances exist for measuring extension, and labels changed only slightly. Moreover, some concepts vanished (e.g. ConditionalEffect, ServiceCategory) and some stayed the same (symmetrical concepts Process, Parameter).

Again, in this scenario, actual drift of meaning can be observed by looking into concept-per-concept stability for intension tables. Observations include that:

- Concepts such as ConditionalEffect and ServiceCategory have vanished, not only because their names are missing, but also their properties (intension) verify this, as no matches are detected. Notably, some complete matches (1.0 highlighted in gray) are to be ignored, as they refer to both concepts having zero properties.
- Concepts such as Parameter, Process, and Input remained the same, as they do have at least one property and their similarity matches the concepts symmetrical in the matrix (highlighted in yellow).
- One concept transformed to another. Precondition and Condition present full stability (highlighted in yellow), meaning that the properties of the former migrated to the latter (in addition to the name similarity which concerns the label aspect).

6.2.3. *Hybrid approach*

The visualized chains for the hybrid approach are shown on Fig. 15. The concepts can easily be classified in two categories: the ones that do not correspond to new concepts (single-node chains) and the ones that entirely transform to new ones (1.0 similarity chains). Conclusions can be made much more easily:

- ConditionalOutput becomes Output (1.0)
- Precondition becomes Condition (1.0)
- Input, Parameter, Process and ServiceProfile stay the same (1.0)

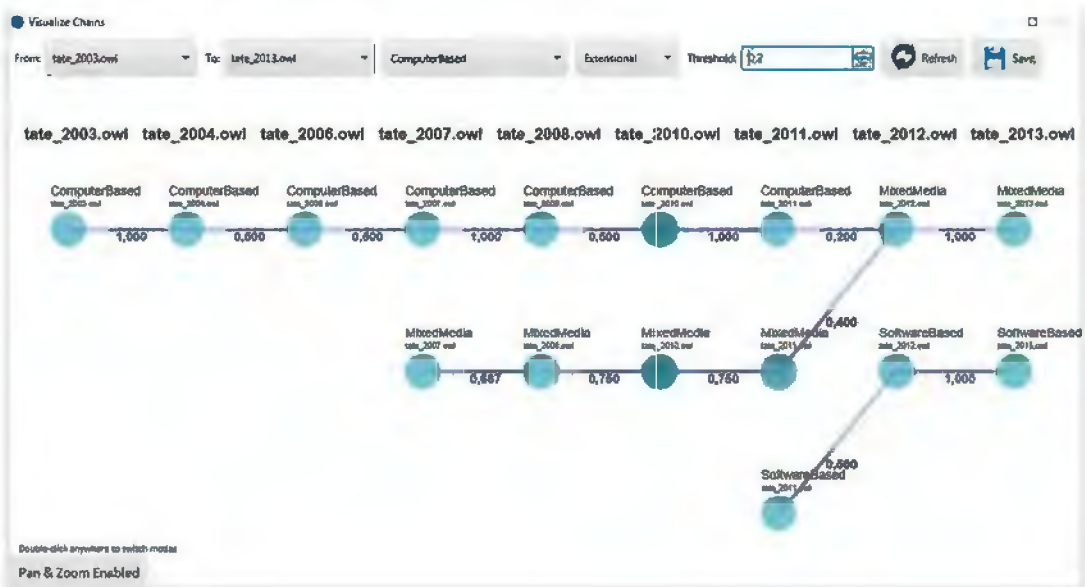


Fig. 13. SemaDriftFx showing extensional aspect hybrid chains for the digital preservation ontology.

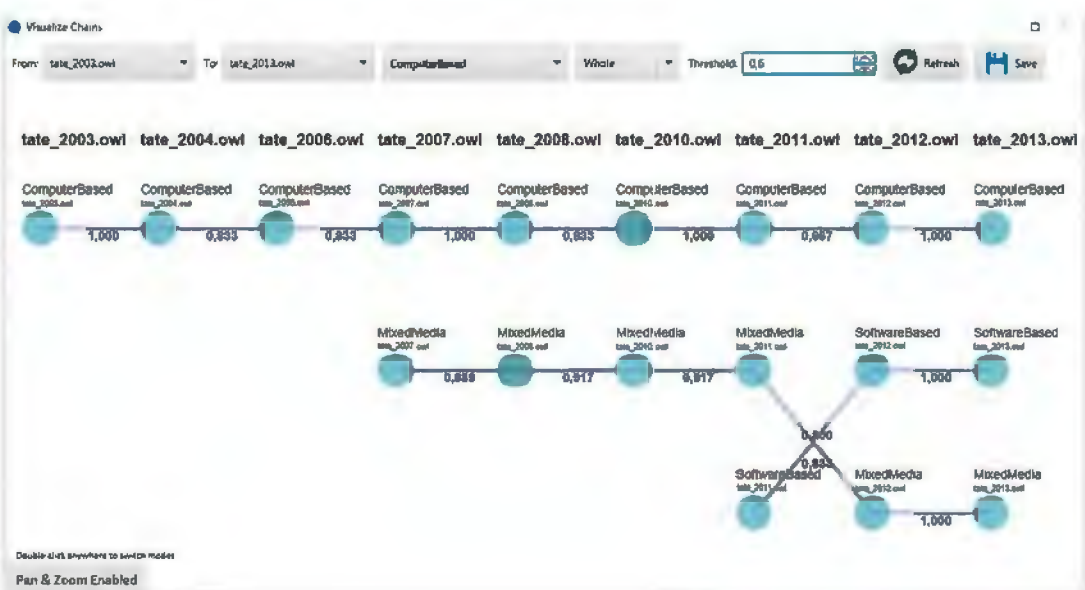


Fig. 14. SemaDriftFx showing whole aspect hybrid chains for the digital preservation ontology.

- Profile loses many of its properties (0.2)
- ConditionalEffect, ServiceCategory, ServiceParameter bear no similarity to new concepts.

7. End-user evaluation

The aim of this section is to give preliminary feedback as to the usability of the tools introduced. For this purpose, an end-user evaluation was conducted for the SemaDriftFx application, central to the Application Suite in terms of functionality and user-friendliness, and allowing participants that were knowledgeable of the Semantic Web technologies but did not necessarily possess capabilities to author ontologies or use Protègè. Another quality that was explored is the amount of prior knowledge needed to comfortably use the application. While this evaluation gives a preliminary sign of usability feedback, a more thorough evaluation methodology is planned for the future for efficiency and effectiveness, using timed tasks and control groups.

The team of users consisted of 26 scholars, who completed PhD studies in the field of informatics or electrical engineering, recruited through direct contact. The users were all knowledgeable in the field of Semantic Web technologies, with the ability to handle ontologies with a GUI, but not necessarily experts in editing the raw format.

In order to motivate the complete exploration of all the functionality offered by the tool prior to evaluation, a predefined set of tasks was given, which explored all functions and options at least once. The tasks they were asked to perform initially focused on learning the tool and then on how to measure and interpret semantic drift. Specifically, the users were provided with the tool and the ontologies of Digital Preservation used in Section 6.1 and they were asked the following:

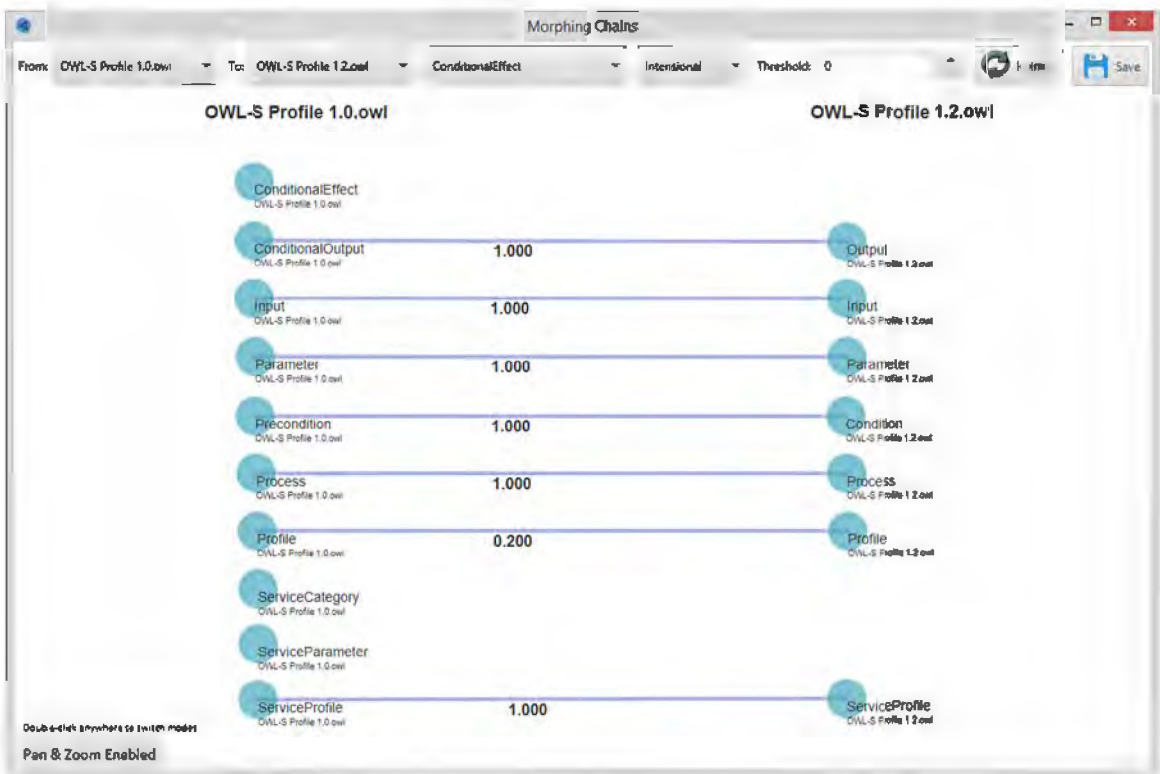


Fig. 15. SemaDriftFx showing hybrid chains for the OWL-S Web Service ontology versions 1.0 and 1.2.

1. Add all the ontologies in random order;
2. Arrange them in chronological order from oldest to newest;
3. Remove one ontology and add it again, rearranging it in chronological order if needed;
4. Using the morphing based approach to measure drift, which aspect do you think is (a) the most stable and (b) the least stable?
5. For the weakest aspect, find a concept-per-concept pair, which indicates a concept drift of splitting meaning between two concepts;
6. Using the hybrid approach which aspect do you think is a) the most stable and b) the least stable.

To evaluate usability, after using the application, the users filled a System Usability Standard (SUS) questionnaire [32]. Table 6 shows the ten fixed SUS questions and some given names of aspects that we consider to relate most to each of them. It can be seen that the statements actually cover a variety of aspects of system usability, such as the need for support, training, and complexity, and thus have a high level of face validity for measuring usability of a system.

The graph in Fig. 16 shows the respective average and standard deviation (sd) of response scores in each area. The evaluation revealed positive user feedback in almost all aspects. The low score on the use frequency, i.e. how often the users imagine they use the application, is the lowest. This could be attributed to the users being Semantic Web experts but not necessarily involved in this field, or to the field's still low penetration in the community. However, the SUS scale items are not meaningful on their own, due to the complementary repetition of some attributes.

The SUS score is equal to the sum of the score contribution from each item. Each item's score contribution will range from 0 to 4. For items 1, 3, 5, 7, and 9 the score contribution is the scale position minus 1. For items 2, 4, 6, 8, and 10 the contribution is 5 minus the scale position. The sum of the scores is then multiplied by 2.5.

Table 6
The SUS questions and the usability aspects most relative to them.

Question	Aspect
I think that I would like to use this system frequently	Use Frequency
I found the system unnecessarily complex	Simplicity
I thought the system was easy to use	Ease-of-use
I think that I would need the support of a technical person to be able to use this system	Independence
I found the various functions in this system were well integrated	Integration
I thought there was too much inconsistency in this system	Consistency
I would imagine that most people would learn to use this system very quickly	Learning Curve
I found the system very cumbersome to use	Efficiency
I felt very confident using the system	Self-Confidence
I needed to learn a lot of things before I could get going with this system	Prior Knowledge

In our case, SUS gives an above-average total score of *80.25, out of 100*, achieved via the high scores in effectiveness aspects and the low scores in prior knowledge required. This outcome points to the need for a more experienced audience or, alternatively, the development of video tutorials, showcases and guidance within the software itself to educate the users about the background of the tool.

8. Discussion

After the methods, applications and use cases are presented, it is clear that there is no optimal solution to every case of scenario and data. Each approach has its own merits and pitfalls that make it suitable for certain cases. The morphing-based approach is the most general one, presenting a concept's similarity with all the concepts of the next ontology for the user to survey. At the same time, that information might easily be overwhelming in

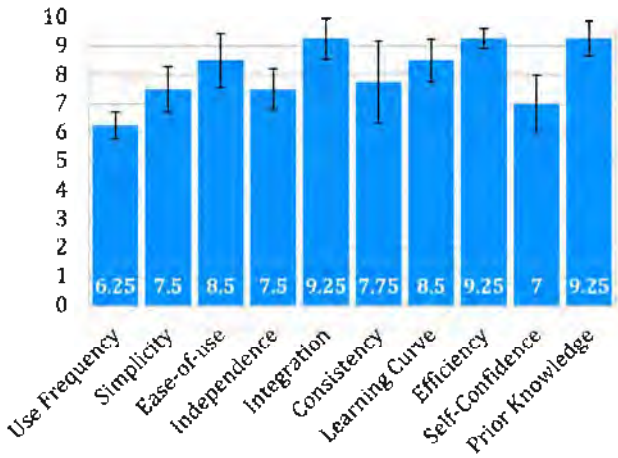


Fig. 16. Average and standard deviation bars of response scores per SUS question/aspect.

large ontologies. The hybrid approach introduced here presents users with an additional means for examining drift. Of course, there is no guarantee that the most similar concept is in fact the corresponding one. But by selecting them, hybrid is able to form chains of concepts and, in turn, rankings. Therefore, it is an ideal choice when large ontologies need to be pruned and examined for semantic drift.

Regarding the methods, an inconclusive issue is grading two empty concepts, i.e. those with no properties (intensional) or no instances (extensional) with 100% similarity. The original method [7] leaves this open to interpretation and our current implementation grades intensional similarity with 100% when both concepts have no properties and extensional when they have no instances. Choosing to grade with 100% instead of 0% favors the method to grab the truly corresponding concepts along with the (possibly many more) false ones. Grading them with 0% would turn them down along with the rest of the empty concepts. The issues assimilates to a problem where lower criteria yield high recall but also low precision.

A solution might be given at the GUI level; after all, the semantic drift methods utterly exist to help the scholars observe and investigate more. The current SemaDrift Protègè Plugin presents the cells of 100% similar but empty concepts, highlighted as gray (as showed in Fig. 5). In contrast, the non-empty 100% similar concepts are more interesting and, thus, highlighted as yellow. This way, the user can take more informed decisions.

9. Conclusions and future work

The SemaDrift Application suite is a collection of tools, methods and metrics to provide the means for a variety of users to measure semantic drift in ontologies in ways previously not possible. The underlying metrics, inspired from the field of Machine Learning are now transferred to the Semantic Web domain and are directly applicable to any chain of ontology versions, regardless of its domain. The methods consider four aspects of drift, label, intension, extension and whole, following a morphing-based approach where concept identity across ontology versions is unknown. At this point, SemaDrift also introduces a novel hybrid approach to pinpoint concept identities without human intervention, by constructing hybrid chains between pairs of concepts with the highest similarity in consecutive versions. With assumed identities known, new metrics and insights such as concept stability ranking are available.

The open-source application suite targets different user types with various tools: a software library for engineers to reproduce and integrate the methods programmatically, a Protégé plugin for ontology authors that are already familiar with the popular software and a stand-alone application with extended functionality to visualize drift for the general user without an expert background. Realistic use cases in digital preservation of art and Web Service semantics have demonstrated that the application as well as the methods can provide valid insights, that would otherwise be much harder to obtain. Semantic Web experts have performed a SUS usability evaluation after performing a series of given tasks, showing positive usability feedback for SemaDriftFx, the most advanced of proposed tools.

Future work begins with a more thorough and methodic evaluation strategy and experiments. The current SUS evaluation presented here merely gives a preliminary sign of usability feedback. In order to gain more insight into the usability but also the usefulness of the hybrid approach itself, a more systematic evaluation is planned. To be exact, we plan to use ontologies from more domains in addition to the ones presented here. In each domain ontology, the users will be assigned with a series of tasks to handle the environment (e.g. open, close, rearrange ontologies), as well as to experiment with the hybrid and morphing-based approach, algorithms and thresholds. The tasks will be timed and will require answers that prove that they have not only performed the tasks, but also managed to identify interesting and meaningful underlying semantic drifts. User groups will be employed to cross-evaluate and benchmark the morphing-based implementation, the hybrid approach and the use of tools against other graphical user interface applications. Other evaluation methods will also be examined, such as PSSUQ,[8] which considers usefulness and efficacy.

Future research directions aim to broaden the scope of domain-independent open tools enabling the Semantic Web community and disseminating the achieved results. So far the core methods developed for calculating drift measures based on ontology evolution have focused on techniques that retain generality without the need for human pre-processing, i.e. annotating identities, such as the morphing-based and the novel hybrid approach. On the other hand, an identity-based method should also be implemented in the future, as it entails far less uncertainty, giving a much clearer picture of concept stability and drift insights. In addition, this could be used as ground-truth to measure the performance of the hybrid approach or observations made via the morphing method. To do so would require user input to indicate the correspondence of a concept across versions, either through metadata input or a graphical user interface to facilitate the process.

The methods themselves can always be enriched with more efficient similarity metrics and optimizing the hybrid method further. As metrics vary, new insights may emerge stemming from limitations — e.g. some metrics for stability may further require normalization. Additionally, certain conditions could trigger additional drift information besides the score. For instance, a similarity due to lack of information, e.g. properties or instances, could cause the score to diminish or those items to be highlighted on the GUI. Provided with an appropriate ground truth, methods can be evaluated. This would also allow a stricter hybrid method with optimized tie-breaking mechanisms and finding exclusive concept identities. The structure of the underlying ontology could also be investigated as to how it affects drift measures. For example, drift in a single concept could be propagated proportionally to its parent and children nodes using spreading activation.

Future work can also expand the use cases to more domains, combined with built-in software tools to pre-process ontologies

[8] http://www.tandfonline.com/doi/abs/10.1080/10447318.2002.9669130.

when needed. Through experience, small changes in ontology format (i.e. syntax) or design may be required for the metrics to work. Such issues can be solved by writing transformation scripts, which may be incorporated into the tool for re-use.

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References

[1] T.G. Stavropoulos, S. Andreadis, E. Kontopoulos, M. Riga, P. Mitzias, I. Kompatsiaris, SemaDrift: A Protégé Plugin for measuring semantic drift in ontologies, in: L. Hollink, S. Darányi, A. Meroño Peñuela, E. Kontopoulos (Eds.), 1st International Workshop on Detection, Representation and Management of Concept Drift in Linked Open Data (Drift-a-LOD) in Conjunction with the 20th International Conference on Knowledge Engineering and Knowledge Management (EKAW), CEUR Workshop Proceedings, Bologna, 2016, pp. 34–41.

[2] M. Klein, D. Fensel, Ontology versioning on the Semantic Web, in: Proc of the Int Semantic Web Working Symposium SWWS 31, 2001, pp. 75–91.

[3] B. Yildiz, Ontology Evolution and Versioning, Vienna University of Technology, Karlsplatz, 2006.

[4] C. Bizer, T. Heath, T. Berners-Lee, Linked data-the story so far, Int. J. Semant. Web Inf. Syst. 5 (3) (2009) 1–22.

[5] P. Pareti, E. Klein, A. Barker, A linked data scalability challenge: Concept reuse leads to semantic decay, in: Proceedings of the ACM Web Science Conference, ACM Press-Association for Computing Machinery, 2016.

[6] L. Stojanovic, A. Maedche, B. Motik, N. Stojanovic, User-driven ontology evolution management, Knowl. Eng. Knowl. Manag.: Ontolog. Semant. Web (2002) 133–140.

[7] S. Wang, S. Schlobach, J. Takens, W. Van Atteveldt, Mapping-chains for studying concept shift in political ontologies, in: CEUR Workshop Proceedings, vol. 551, 2009, pp. 13–24.

[8] T.G. Stavropoulos, S. Andreadis, M. Riga, E. Kontopoulos, P. Mitzias, I. Kompatsiaris, A framework for measuring semantic drift in ontologies, in: 1st Int. Workshop on Semantic Change & Evolving Semantics, SuCCESS'16, CEUR Workshop Proceedings, vol. 1824, Leipzig, Germany, 2016.

[9] T.G. Stavropoulos, E. Kontopoulos, A. Meroño-Peñuela, I. Kompatsiaris, Cross-domain semantic drift measurement in ontologies using the semadrift tool and metrics, in: 3rd Workshop on Managing the Evolution and Preservation of the Data Web, MEPDaW 2017, CEUR Workshop Proceedings, vol. 1824, Portoroz, Slovenia, 2017.

[10] P. Wittek, S. Darányi, E. Kontopoulos, T. Moysiadis, I. Kompatsiaris, Monitoring term drift based on semantic consistency in an evolving vector field, in: Proceedings of the International Joint Conference on Neural Networks, 2015-Septe, IEEE, 2015, pp. 1–8.

[11] J. Gulla, G. Solskinnsbakk, P. Myrseth, Semantic drift in ontologies, in: Proceedings of 6th International Conference on Web Information Systems and Technologies, WEBIST, Valencia, Spain, vol. 2, 2010, pp. 13–20.

[12] M. Tury, M. Bieliková, An approach to detection ontology changes, in: Workshop Proceedings of the Sixth International Conference on Web Engineering - ICWE '06, ACM, 2006, p. 14.

[13] S. Wang, S. Schlobach, M. Klein, Concept drift and how to identify it, J. Web Semant. 9 (3) (2011) 247–265.

[14] A. Meroño-Peñuela, C. Guéret, R. Hoekstra, S. Schlobach, Detecting and reporting extensional concept drift in statistical linked data, in: CEUR Workshop Proceedings, vol. 1549, 2013.

[15] M. Uschold, Creating, integrating and maintaining local and global ontologies, in: Proceedings of the First Workshop on Ontology Learning (OL-2000) in Conjunction with the 14th European Conference on Artificial Intelligence, ECAI 2000, Berling, Germany, 2001.

[16] S. Klarman, R. Hoekstra, M. Bron, Versions and applicability of concept definitions in legal ontologies? in: CEUR Workshop Proceedings, 496, 2009.

[17] N. Fanizzi, C. Amato, F. Esposito, Conceptual clustering: Concept formation, drift and novelty detection, in: The Semantic Web: Research and Applications, 5th European Semantic Web Conference, ESWC 2008, Tenerife, Canary Islands, Spain, June 1–5, 2008, Proceedings, Springer, 2008.

[18] W.L. Hamilton, J. Leskovec, D. Jurafsky, Diachronic word embeddings reveal statistical laws of semantic change, 2016. arXiv preprint arXiv:1605.09096.

[19] L. Frermann, M. Lapata, A bayesian model of diachronic meaning change, Trans. Assoc. Comput. Linguist. 4 (2016) 31–45.

[20] N. Guarino, C. Welty, A formal ontology of properties, in: International Conference on Knowledge Engineering and Knowledge Management, Springer, 2000, pp. 97–112.

[21] A. Gangemi, N. Guarino, C. Masolo, A. Oltramari, L. Schneider, Sweetening ontologies with DOLCE, in: International Conference on Knowledge Engineering and Knowledge Management, Springer Berlin Heidelberg, 2002, pp. 166–181.

[22] A. Meroño-Peñuela, R. Hoekstra, What is linked historical data? in: International Conference on Knowledge Engineering and Knowledge Management, Springer, 2014, pp. 282–287.

[23] J. Euzenat, P. Shvaiko, et al., Ontology Matching. Vol. 18, Springer, 2007.

[24] T. Stavropoulos, S. Andreadis, N. Bassiliades, D. Vrakas, I. Vlahavas, The tomaco hybrid matching framework for SAWSDL Semantic Web Services, IEEE Trans. Serv. Comput. 9 (6) (2016) 954–967.

[25] A.E. Monge, C. Elkan, The field matching problem: Algorithms and applications, in: 2nd Intl. Conf. Knowledge Discovery and Data Mining, KDD, 1996, pp. 267–270.

[26] J.H. Gennari, M.A. Musen, R.W. Fergerson, W.E. Grosso, M. Crubézy, H. Eriksson, N.F. Noy, S.W. Tu, The evolution of Protégé: an environment for knowledge-based systems development, Int. J. Human-Comput. Stud. 58 (1) (2003) 89–123.

[27] M. Horridge, S. Bechhofer, The owl api: A java api for owl ontologies, Semantic Web 2 (1) (2011) 11–21.

[28] S. Ross, Digital preservation, archival science and methodological foundations for digital libraries, New Rev. Inf. Netw. 17 (1) (2012) 43–68.

[29] N. Lagos, E. Kontopoulos, M. Riga, P. Mitzias, G. Meditskos, S. Waddington, P. Laurenson, Designing for inconsistencythe dependency-based PERICLES approach, in: East European Conference on Advances in Databases and Information Systems, Springer International Publishing, 2015, pp. 458–467.

[30] P. Mitzias, M. Riga, S. Waddington, E. Kontopoulos, G. Meditskos, P. Laurenson, I. Kompatsiaris, An ontology design pattern for digital video, in: Proceedings of the 6th Workshop on Ontology and Semantic Web Patterns, WOP 2015, 2015.

[31] E. Kontopoulos, M. Riga, P. Mitzias, S. Andreadis, T. Stavropoulos, N. Lagos, J.-Y. Vion-Dury, G. Meditskos, P. Falcão, P. Laurenson, I. Kompatsiaris, Ontology-based Representation of Context of Use in Digital Preservation, in: 1st Workshop on Humanities in the Semantic Web - WHiSe, Co-Located with the 13th Extended Semantic Web Conference, ESWC 2016, At Heraklion, Crete, Greece, in: CEUR Workshop Proceedings, vol. 1608, 2016, pp. 65–72.

[32] J. Brooke, et al., SUS-A quick and dirty usability scale, Usability Eval. Ind. 189 (194) (1996) 4–7.